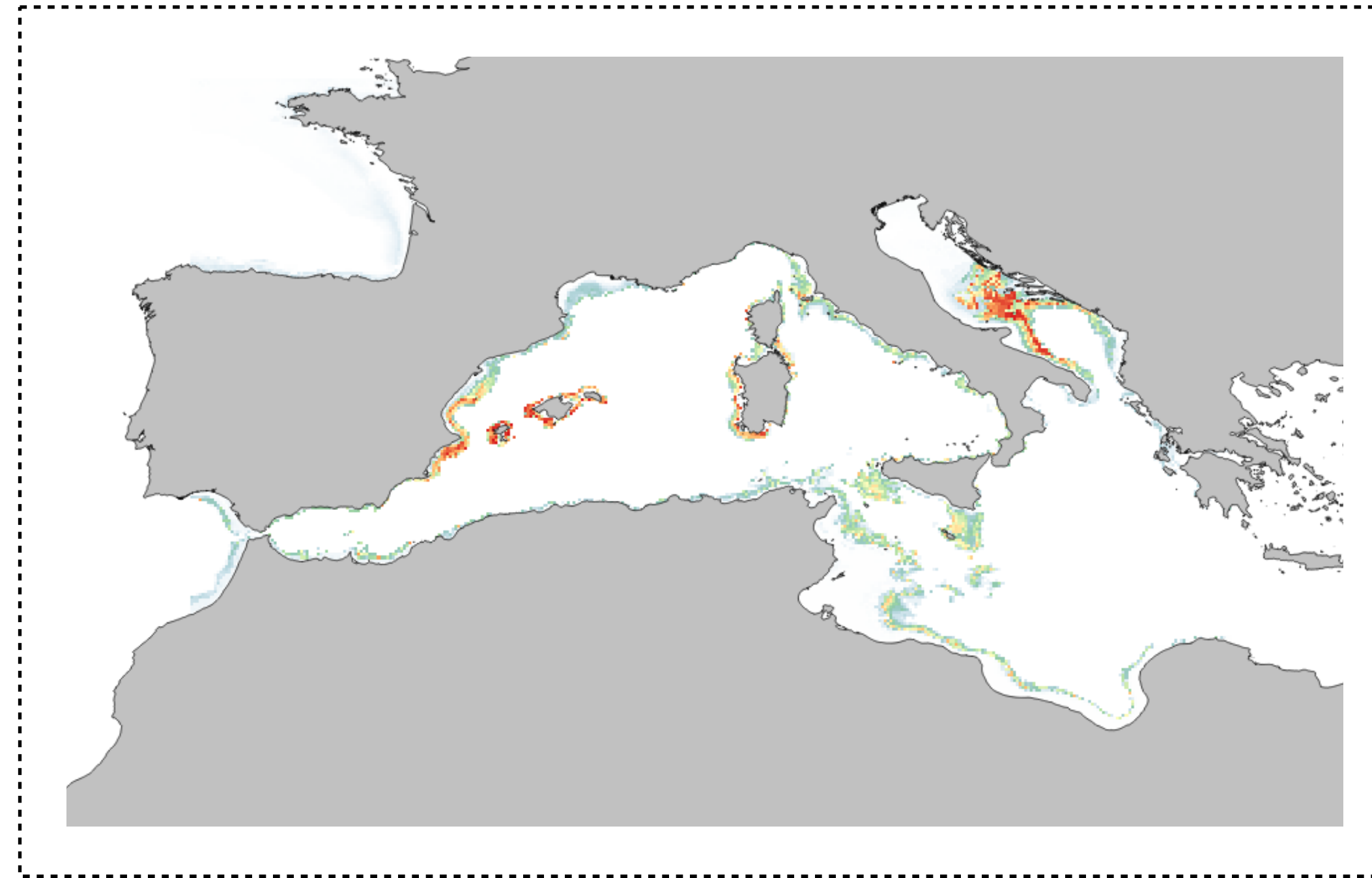


Evaluating, Transferring, and Ensembling

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Measuring SDM Performance



Model evaluation / performance

Verify consistency: do predictions match observations (presences and absences)?

Assess transferability: how well does the model generalize to new data?

Evaluate ecological realism: does the model make biological sense?

Fundamental question: is the model suitable for the purpose (e.g., impact assessment)?

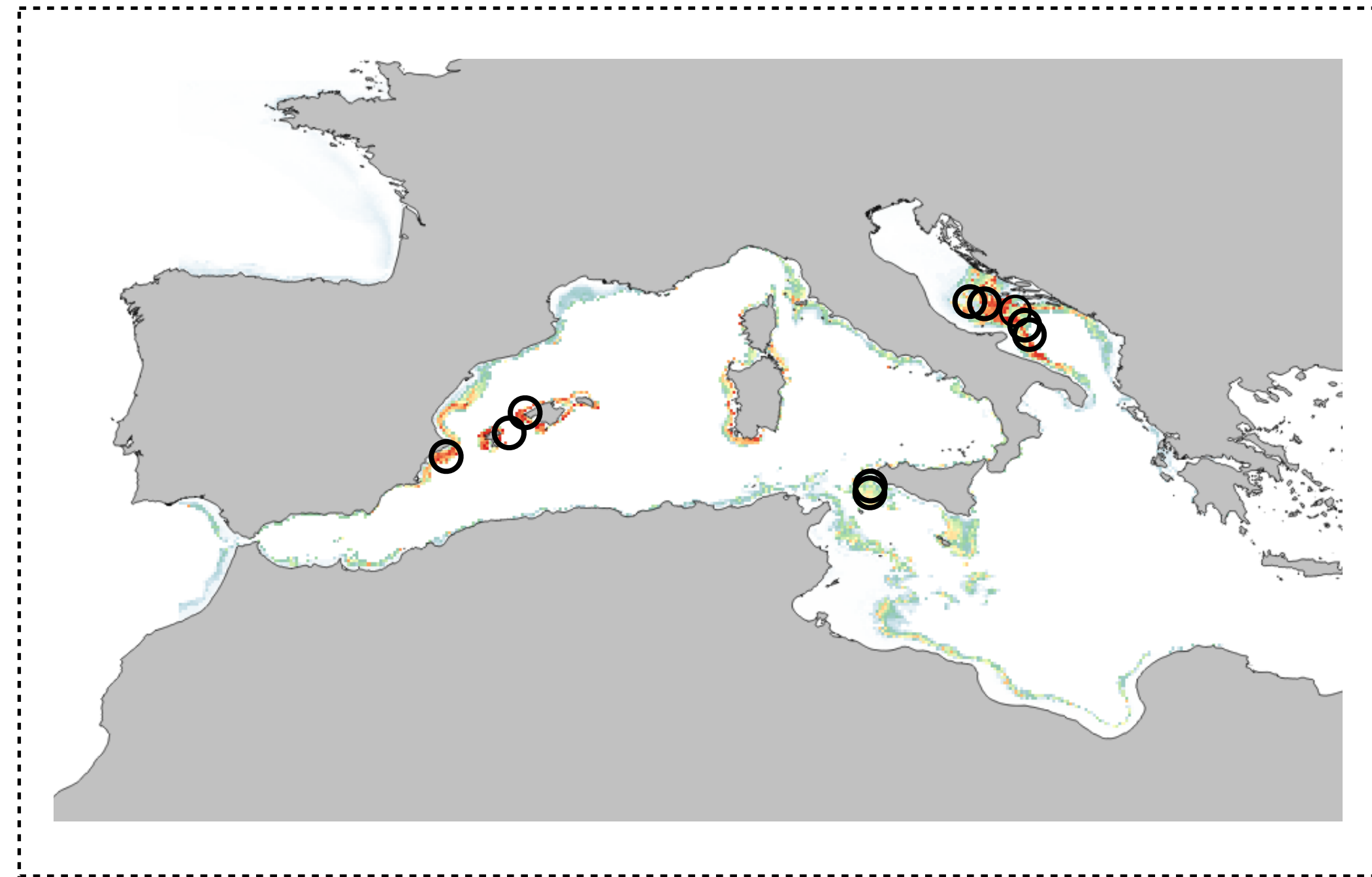
Model evaluation

Comparing predictions with observations we assess for '**false positives**' - model predicts occurrence where the species is absent - and '**false negatives**' - model predicts absence where the species is present; can be summarized in confusion matrices.

Confusion matrix Types of prediction errors		Observation	
		Presence	Absence
Prediction	Presence	True Positive	False Positive
	Absence	False Negative	True Negative

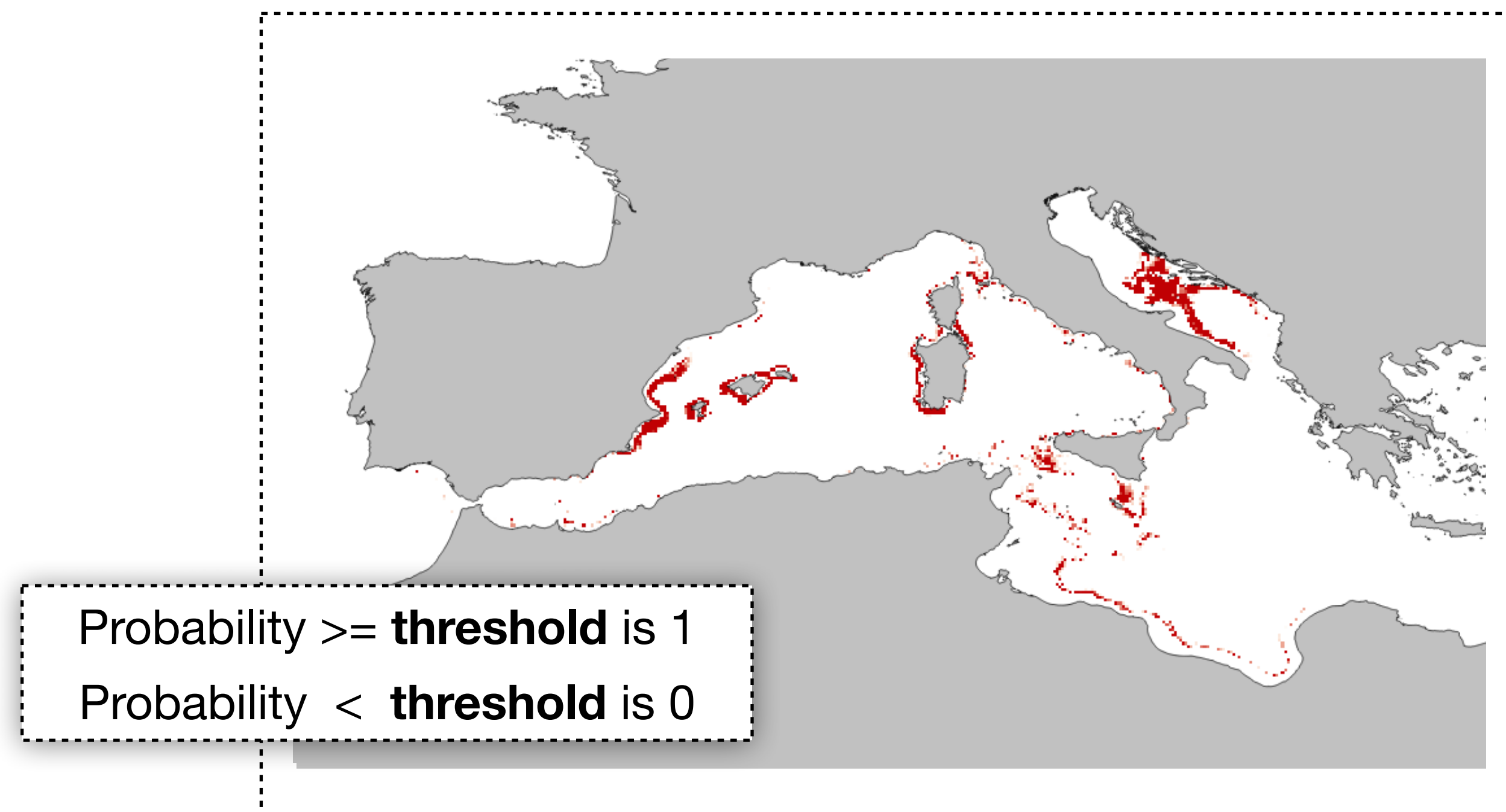
Perfect models only retrieve true positives and true negatives.

Predictions are continuous surfaces (e.g., probability, from 0 to 1).
One cannot compare observations (e.g., 1s) with the corresponding model output (e.g., $P: 0.7$).



Predictions are continuous surfaces (e.g., probability, from 0 to 1).

One cannot compare observations (e.g., 1s) with the corresponding model output (e.g., $P: 0.7$).



Predictions need to be reclassified as binary responses (presence-absence) for comparison with the observed data.

Evaluation metrics: threshold-dependent

The elements of the confusion matrix are used to compute various evaluation criteria that measure the performance of the model, **after applying a threshold to predictions**:

Sensitivity: proportion of presences correctly predicted, i.e., true positives;

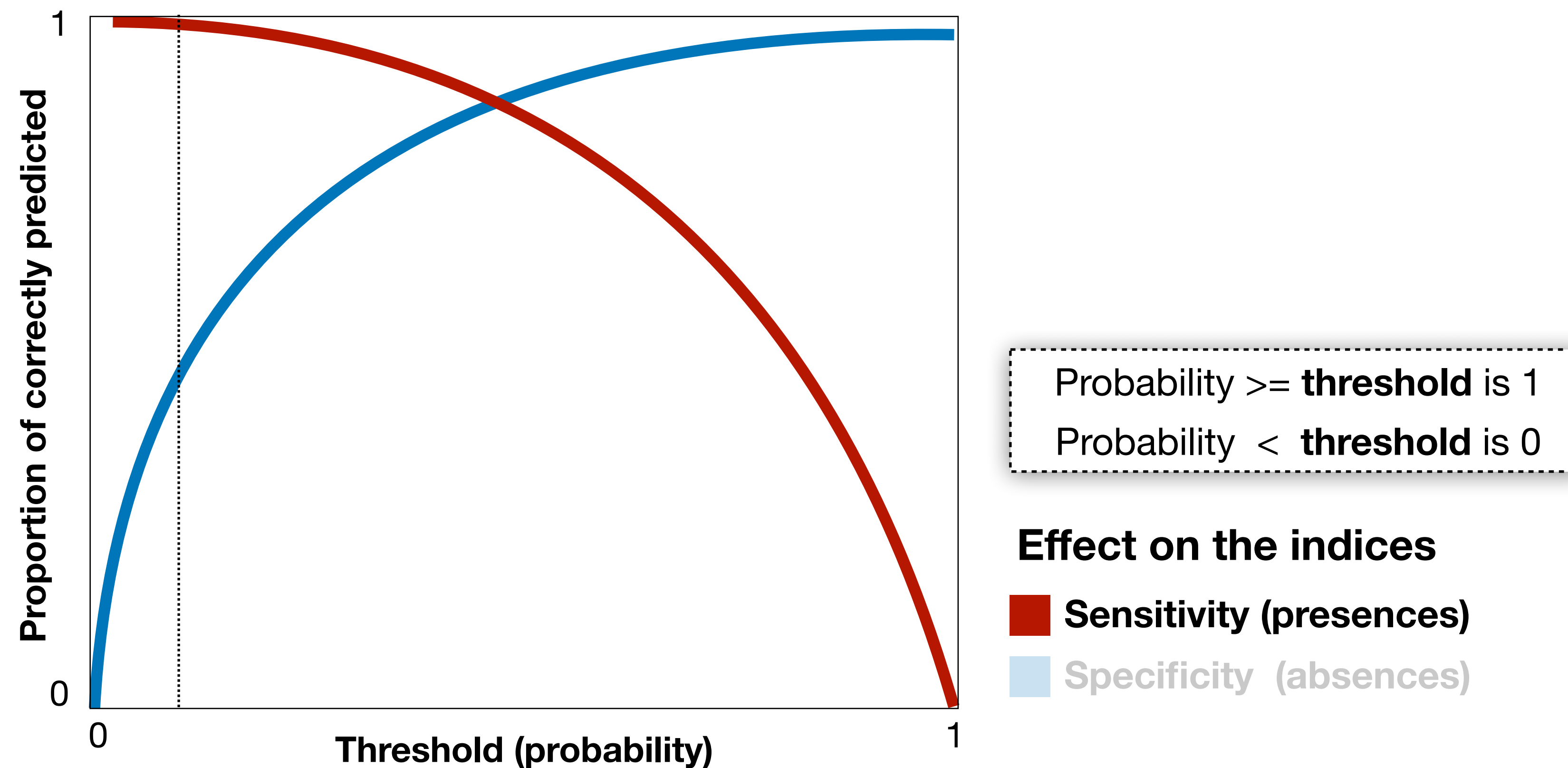
Specificity: proportion of absences correctly predicted, i.e., true negatives;

True Skill Statistics: $\text{Sensitivity} + \text{Specificity} - 1$

TSS describes how well the model predicts presences and absences;

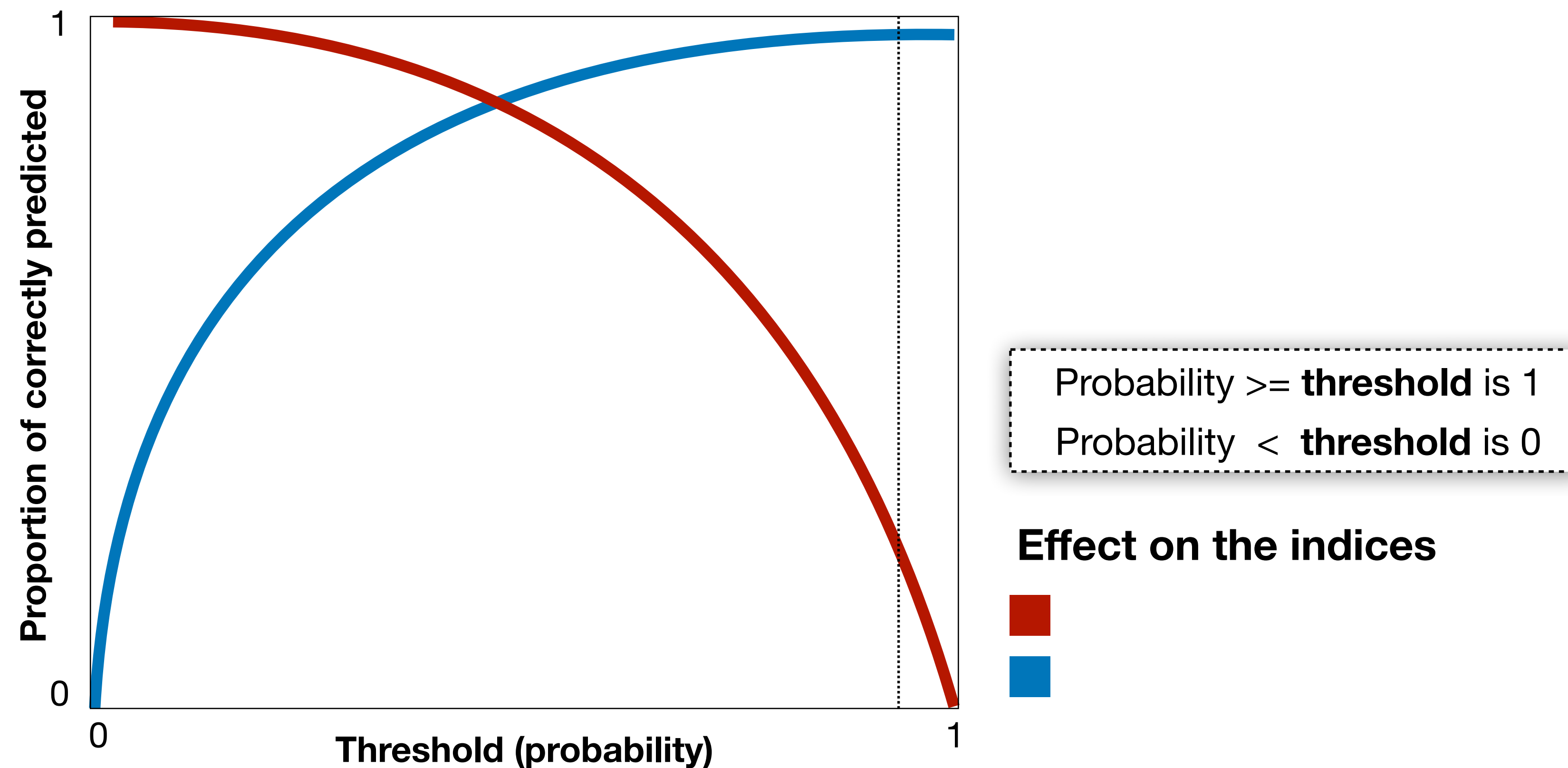
TSS ranges from -1 to +1 (0 $\leq$ no better than a random model).

Sensitivity and specificity vary with the range of thresholds (0 to 1) used to reclassify the models to binary outputs.



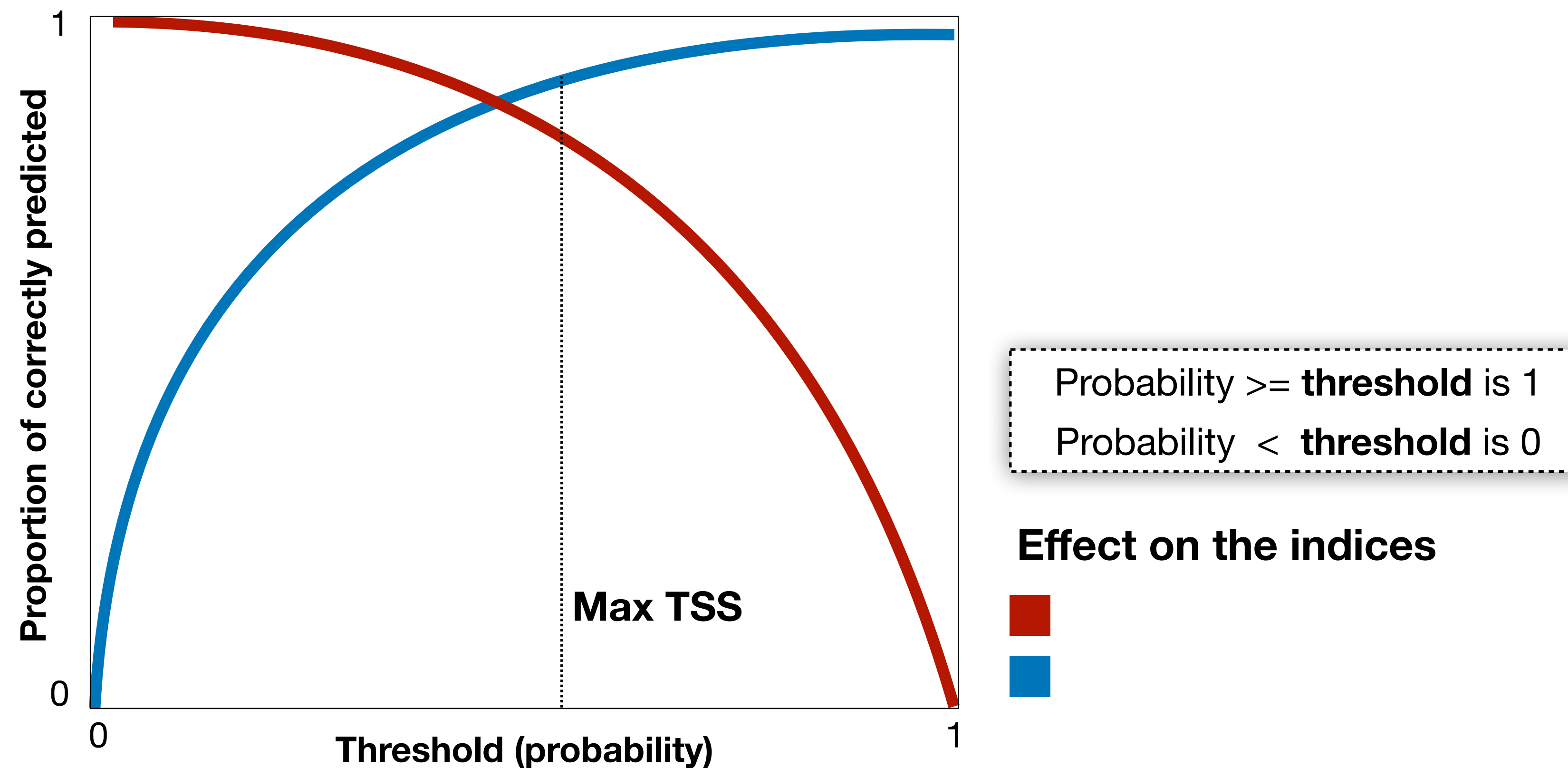
Using a low threshold most cells (in the map) will return 1.

Sensitivity and specificity vary with the range of thresholds (0 to 1) used to reclassify the models to binary outputs.

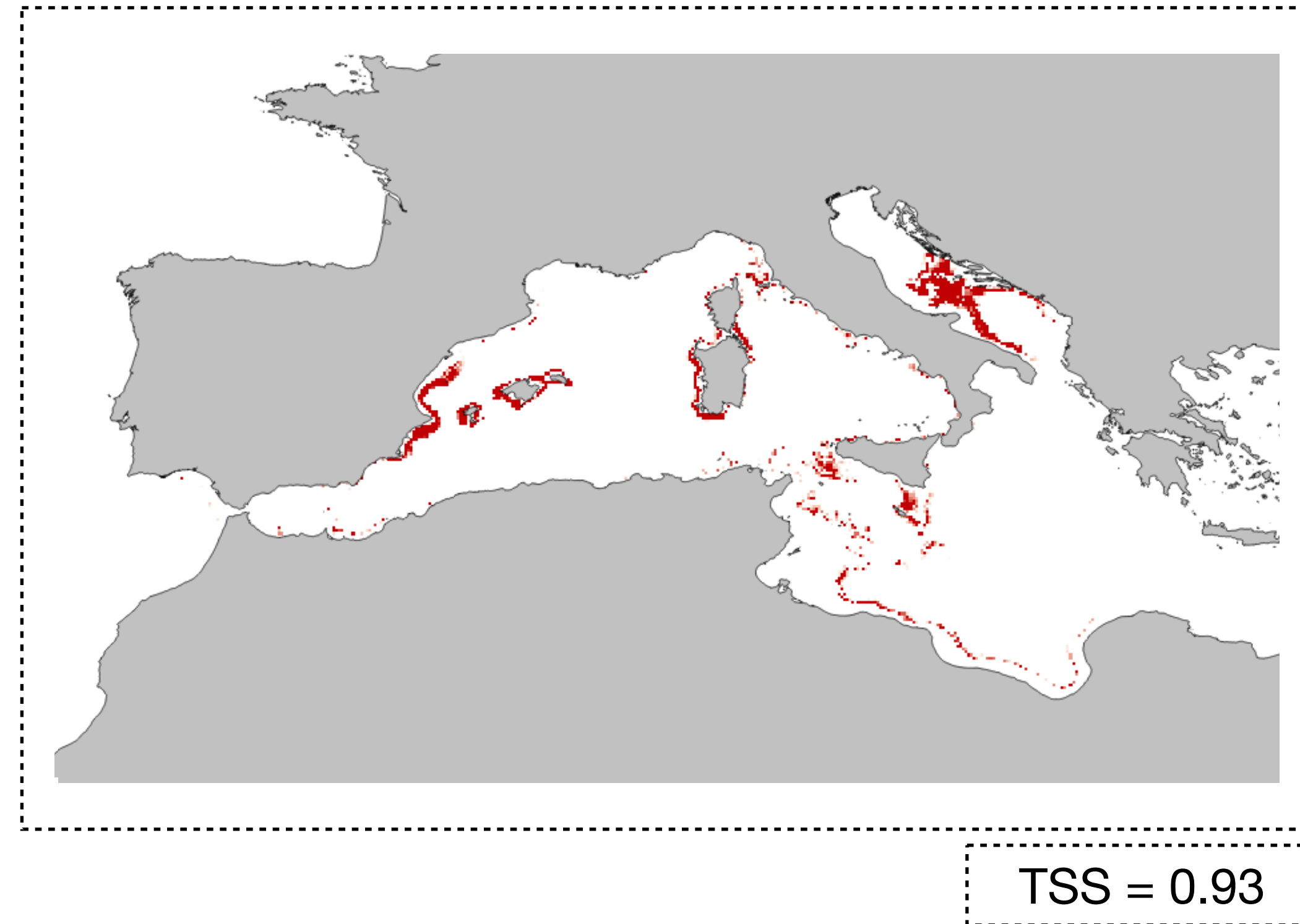


Using a high threshold most cells (in the map) will return 0.

There are **threshold rules to maximize the agreement** between observed data and the predicted binary surface.

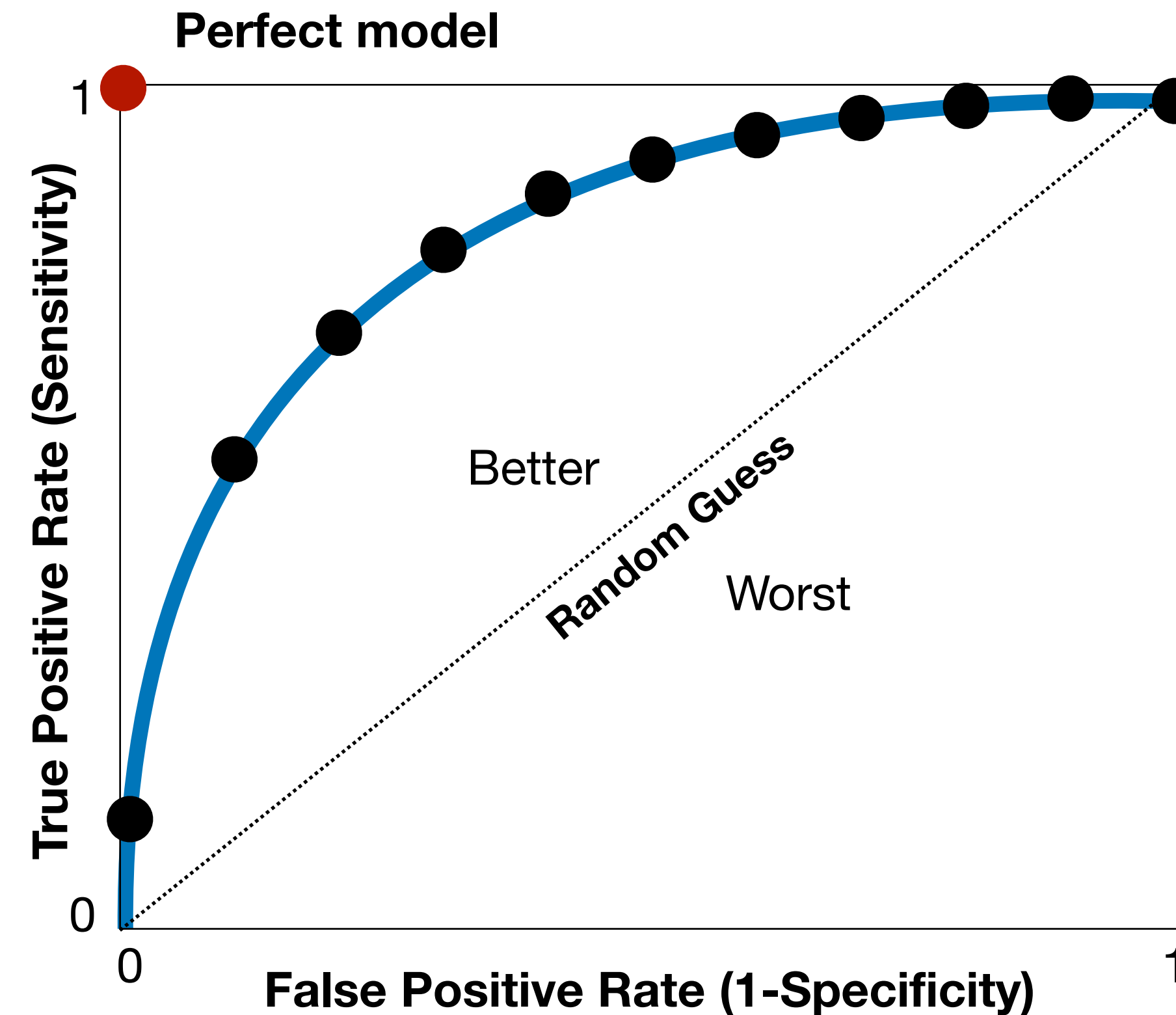


(!) **Maximization of sensitivity + specificity**



Threshold-dependent measures of accuracy

Thresholds like “Max. TSS” allow **assessing the agreement between observed data and the model output** and **converting probability maps into binary maps**.



Threshold-independent evaluation criteria

Area Under the Curve of the Receiver Operating Characteristic Plot.

Compared **true positive rate (sensitivity)** against **false positive rate (1-specificity)** across the range of all possible thresholds. The closer the curve from y-axis, the larger the **AUC**, and thus the more accurate the model.

AUC performance

Accuracy indices like AUC can be interpreted as follow:

- 1 - 0.9 : excellent model
- 0.9 - 0.8 : good model
- 0.8 - 0.7 : fair model
- 0.5 - 0.7 : poor model
- 0 - 0.5 : random model

The Flaw of ROC-AUC: While the ROC-AUC is the most popular metric, it can be highly **misleading when dealing with imbalanced datasets** - a very common scenario in SDMs where you might have 100 presence records and 10,000 background points.

Why it Fails: ROC-AUC **relies on Specificity**. With a massive amount of background points, **the impact of few occurrence records (presences) wrongly classified won't significantly impact the score**, artificially inflating the model's perceived performance.

To solve this, we use the **Precision-Recall Curve** and the **Boyce index**, which completely **ignore Specificity and focuses on Sensitivity**.

The Area Under the Precision-Recall Gain Curve (AUPRG): When the model predicts a presence, how often is it correct? Out of all presences, how many did the model successfully predicted?

Range 0 to 1 (0 = no better than a random model).

The Boyce index: Tests (by correlation) if higher modeled suitability corresponds to higher densities of presence points, which is the desired model behavior.

Range -1 to +1 (0 <= no better than a random model).

Model evaluation

Multi-faceted approach: don't rely on a single metric.

Use a suite:

Sensitivity;

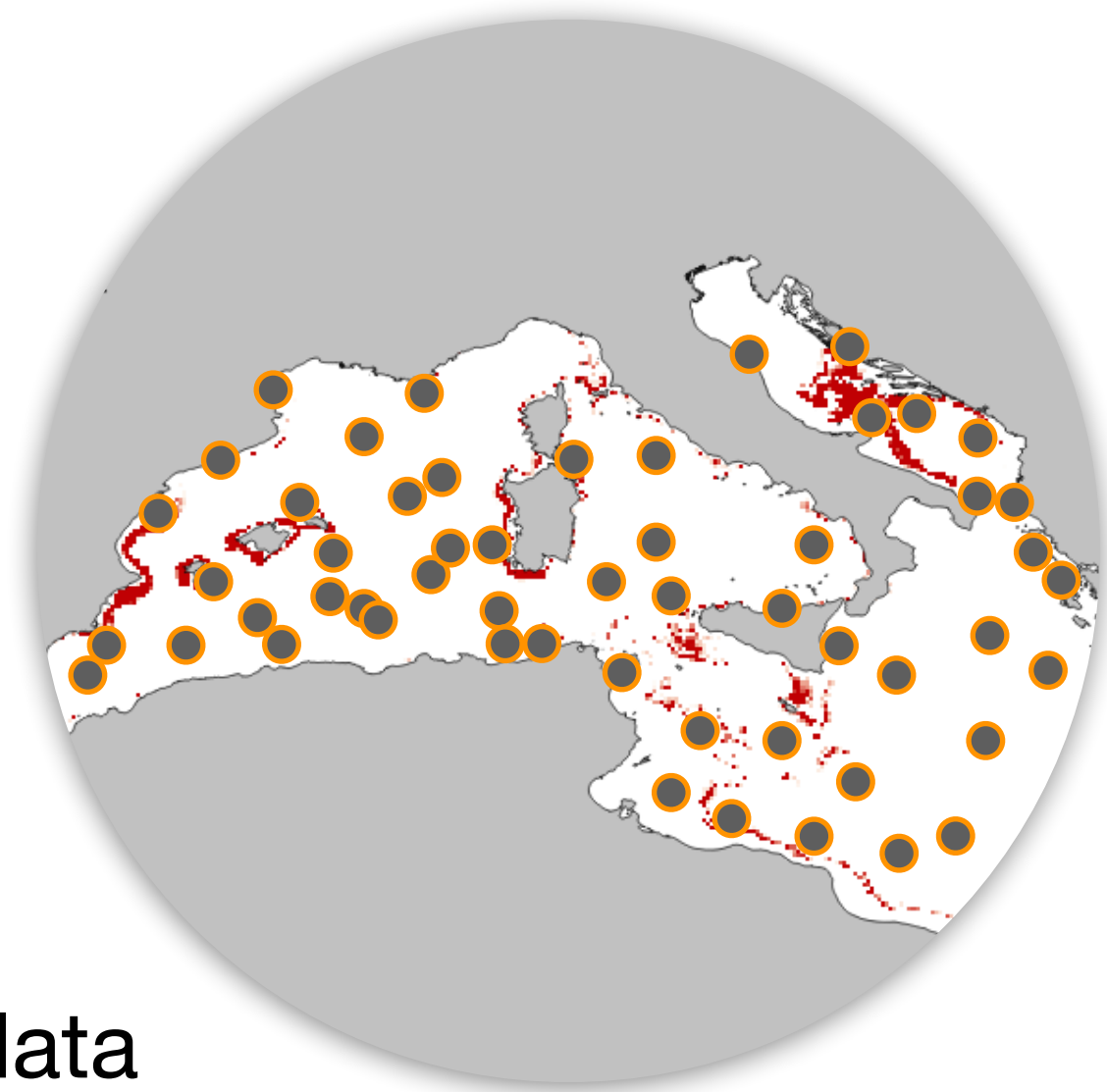
AUC-PRG;

Boyce;

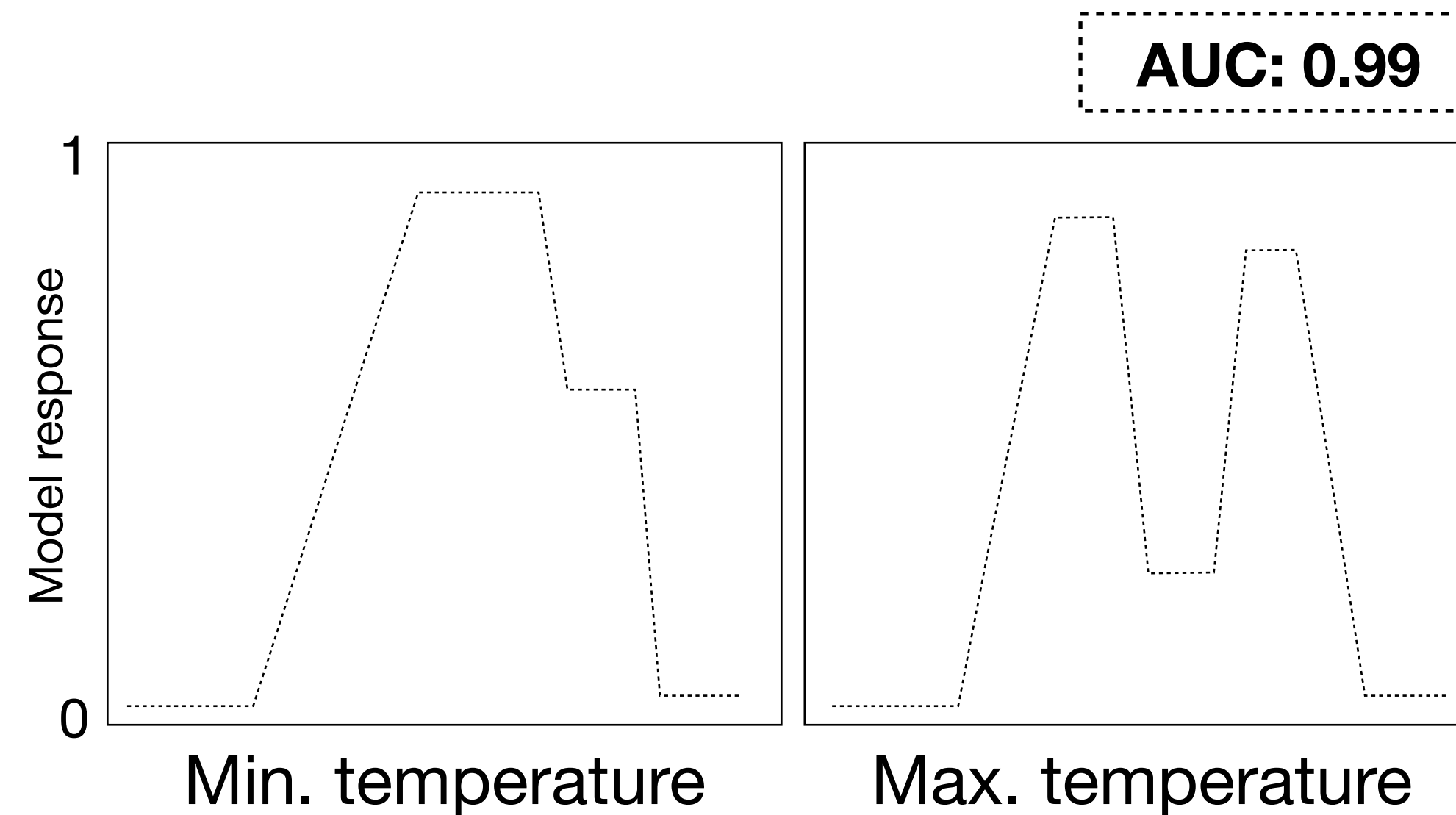
Rigorous Cross-Validation Strategies

High performance linked to good model transferability?

Depends on how it is measured.



- Testing data
- Training data

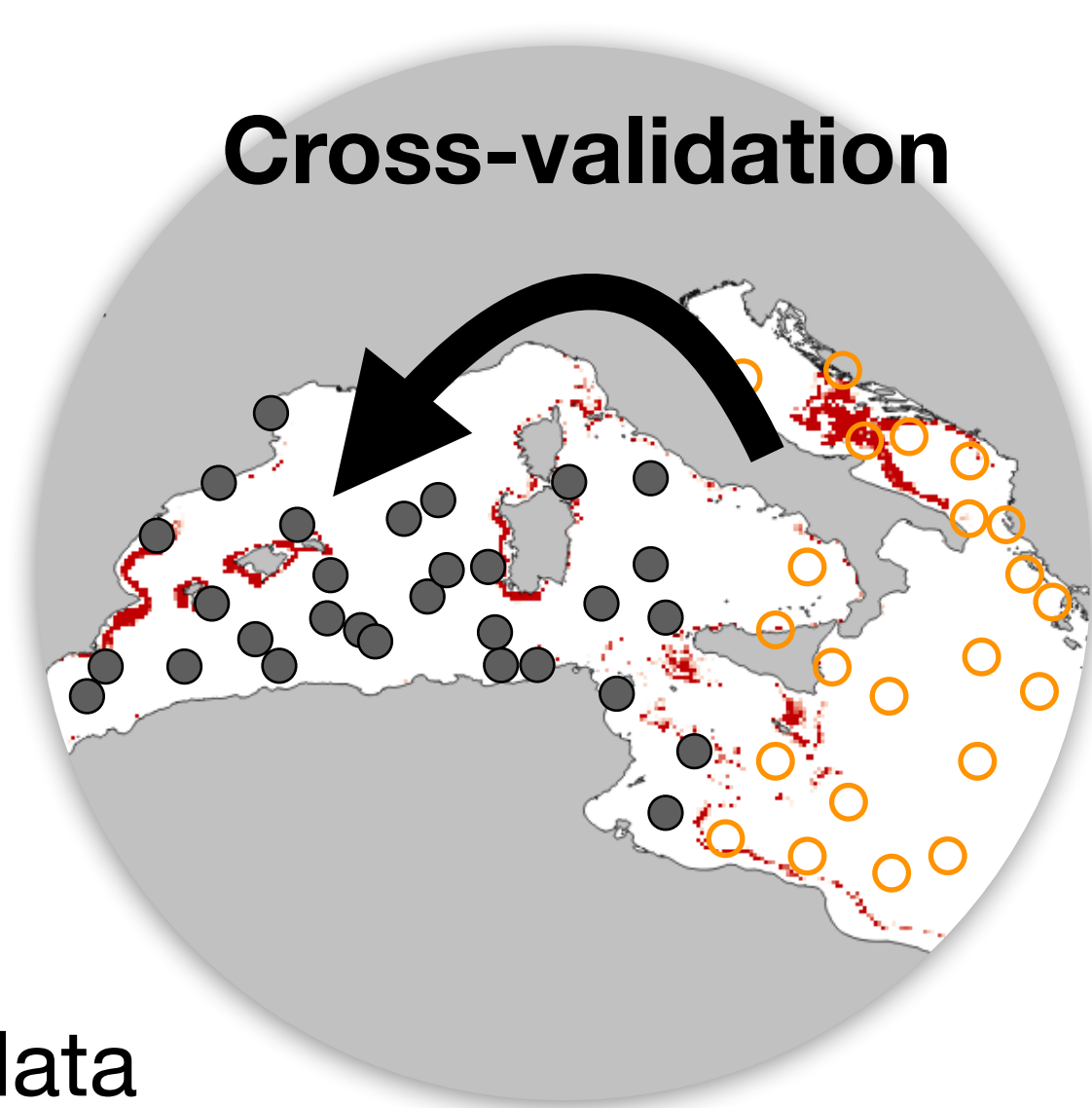


Testing performance with the training data leads to an overestimation of accuracy, regardless of the model's potential for transferability or if it ecologically sound. If the model is largely overfitted - the test is performed in the locations of train - no way to assess for generality.

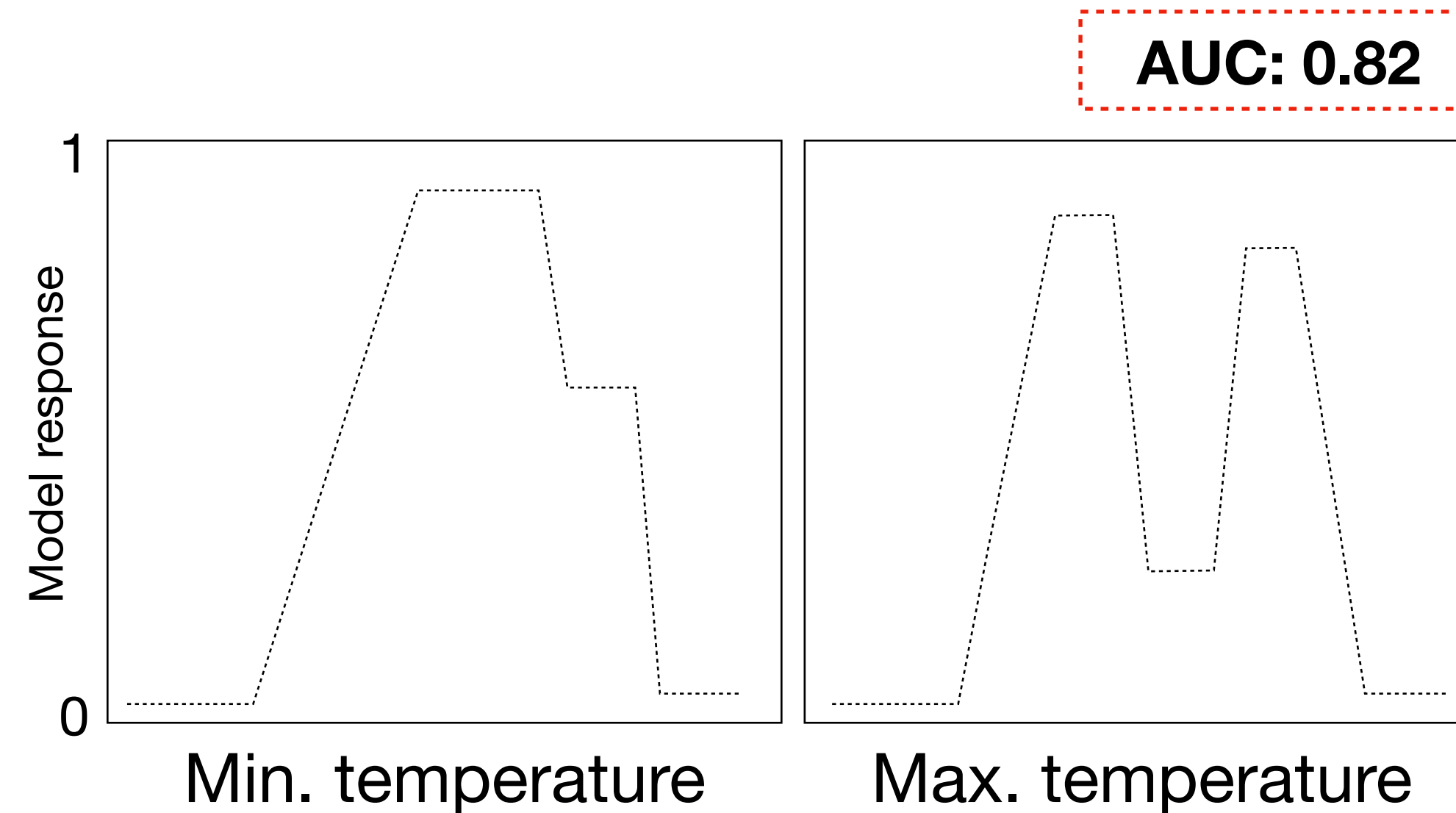
Rigorous Cross-Validation Strategies

High performance linked to good model transferability?

Depends on how it is measured.



- Testing data
- Training data



Testing accuracy with independent data is the approach to evaluate if the model generalized the relationship between occurrences and environment, and its transferability.

This is the same as projecting to other places or times.
Tends to lead to lower accuracy scores, but more reliable scores.

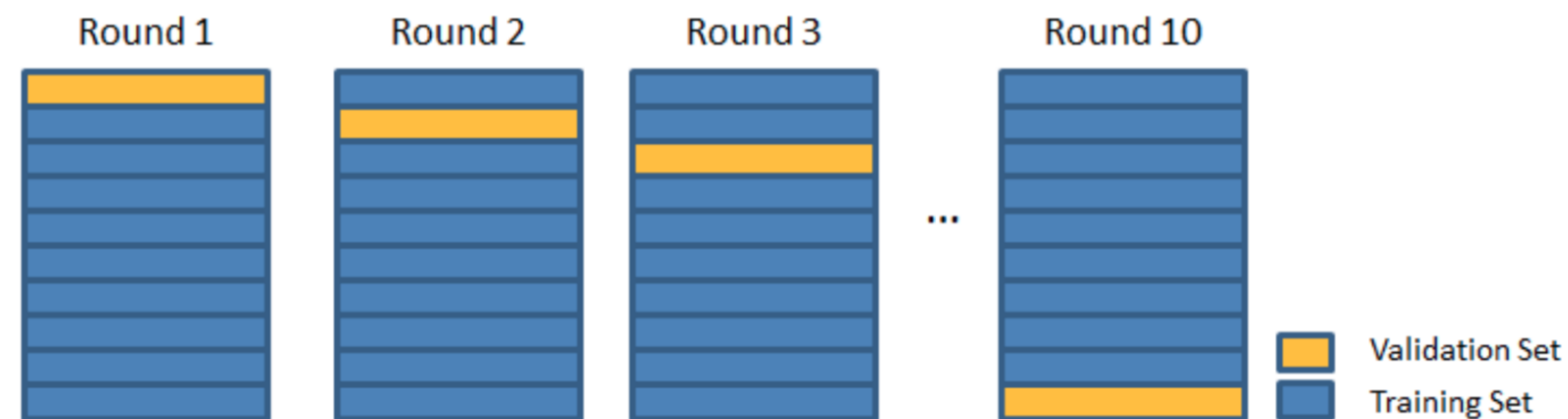
Missing independent data?

Problem: Niche truncation if occurrence data is yielded for testing.

Approach: partitioning the data in k cross-validation interactions, with data splits k times, yielding k estimates of accuracy that can be averaged.

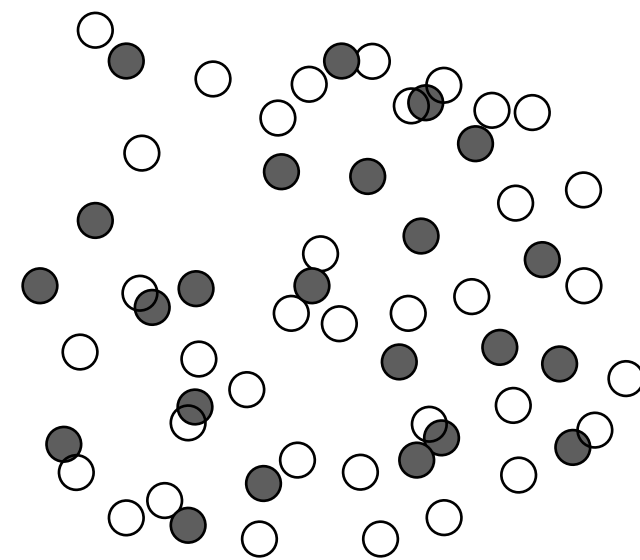
e.g.,

In 10-fold CV, 9/10 of the observations are **used to train the model** and the **remaining 1/10** are **used to estimate performance**; this is repeated ten times and the estimated performance measures are averaged.



Missing independent data

There are different methods to produce independent datasets. **Some approaches provide more independent datasets than others.**



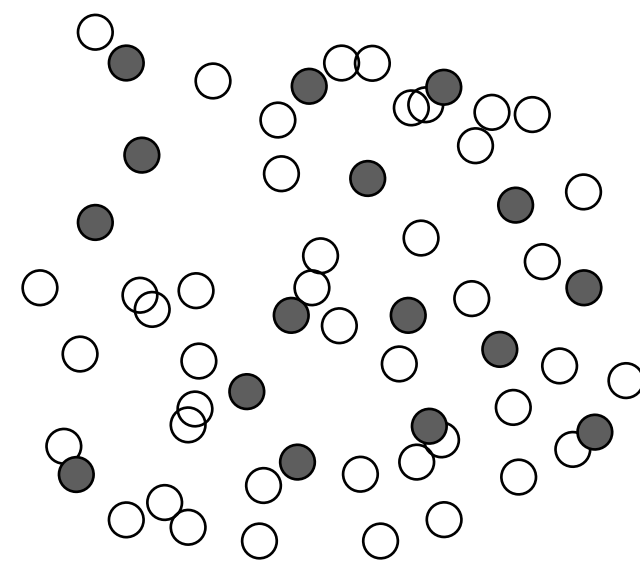
Random (70/30)

(70/30 | k-fold)

○ Training data ● Testing data

Missing independent data

There are different methods to produce independent datasets. **Some approaches provide more independent datasets than others.**



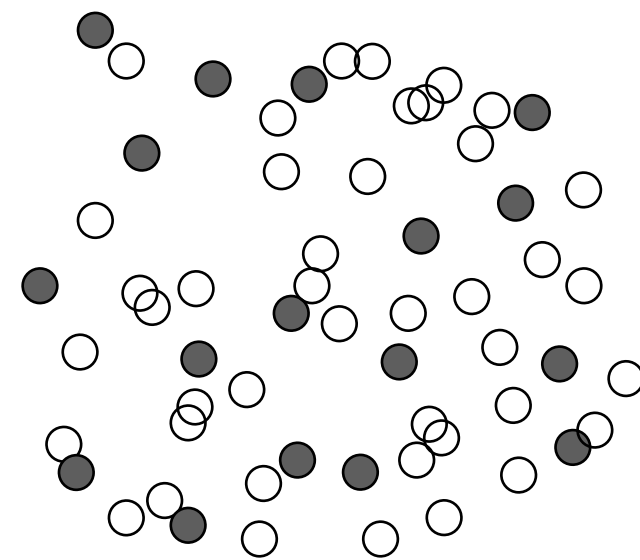
Random (70/30)

(70/30 | k-fold)

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There are different methods to produce independent datasets. **Some approaches provide more independent datasets than others.**



Random (70/30)

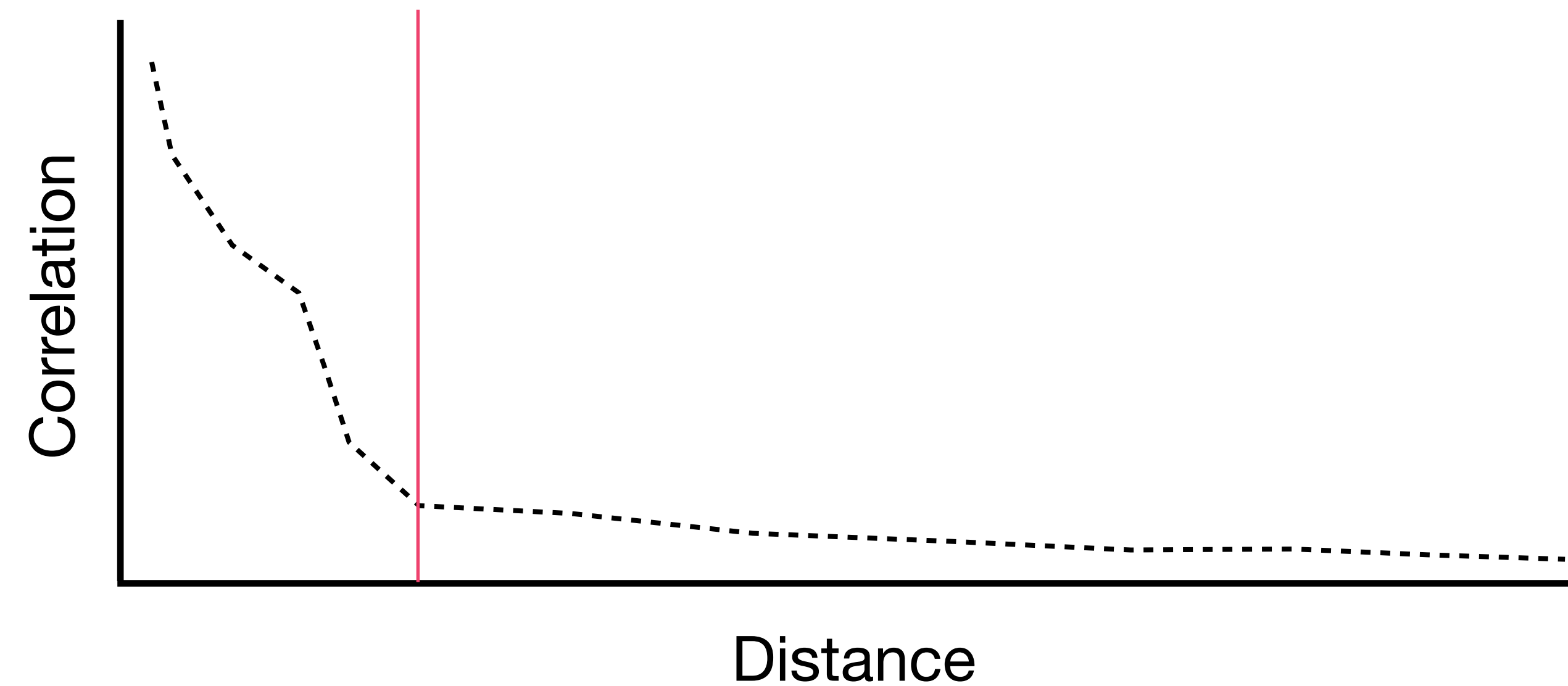
(70/30 | k-fold)

○ Training data ● Testing data

In standard (random) k-fold CV, train and test data are at nearby locations - often not independent.

Spatial autocorrelation: *"Near things are more related than distant things".*

Can **overestimate performance** because training and testing folds contain similar data. **Violates the independence assumption** of statistical tests.



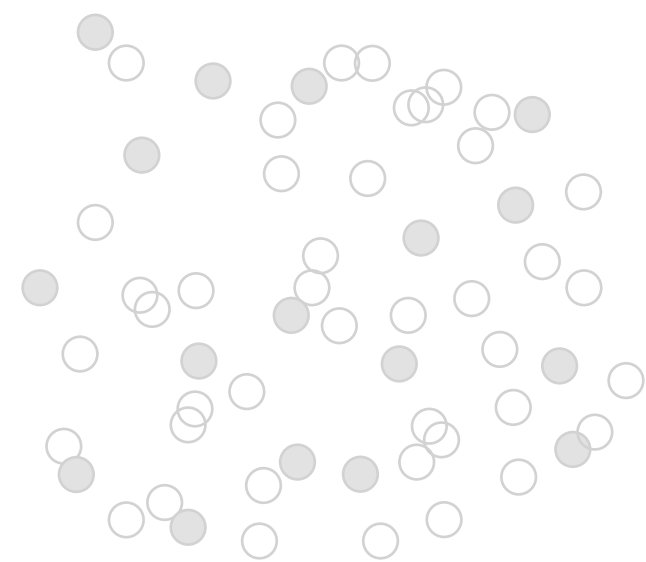
The consideration of Spatial Autocorrelation (SAC)

Records close to one another share nearly identical environmental conditions.

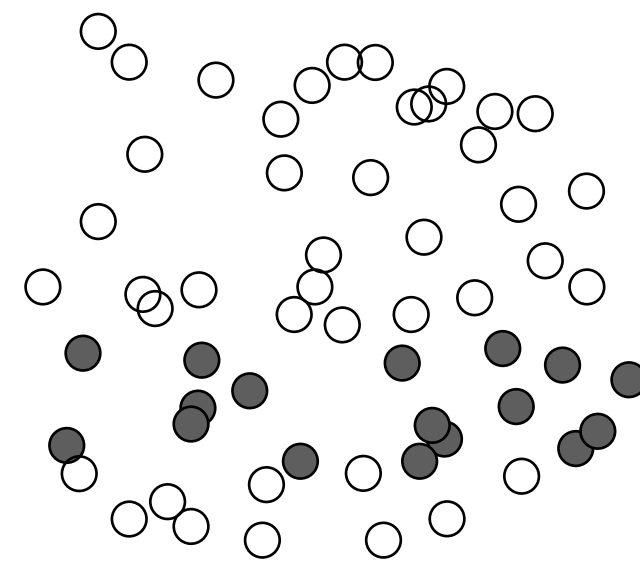
We estimate the distance at which records are not SAC - to define independence.

Missing independent data

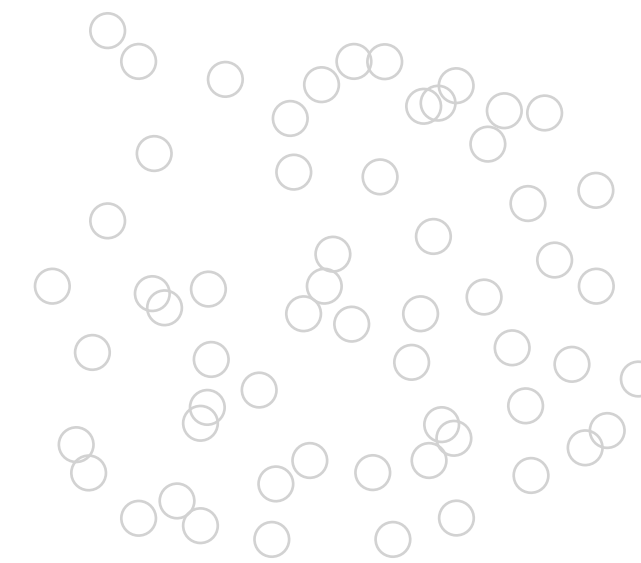
There are different methods to produce independent datasets. **Some approaches provide more independent datasets than others.**



Random (70/30)
(70/30 | k-fold)



Bands
(latitudinal, longitudinal)

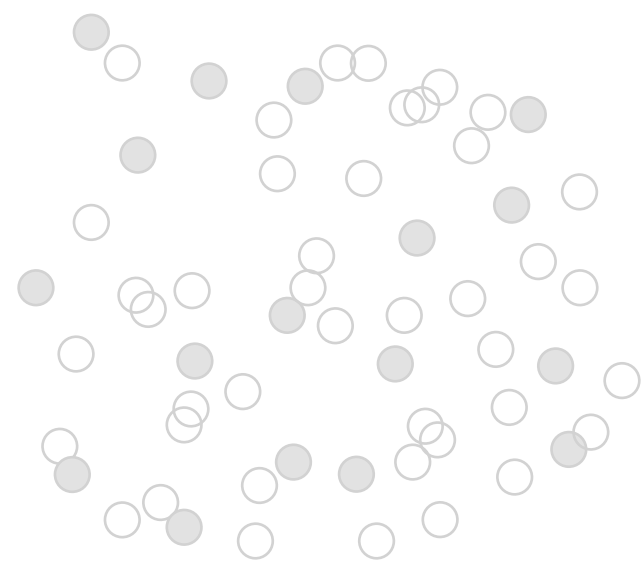


Blocks
(latitudinal, longitudinal)

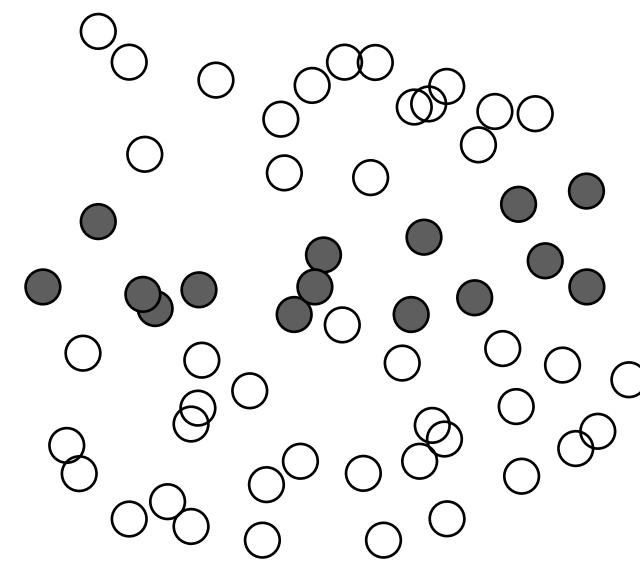
○ Training data ● Testing data

Missing independent data

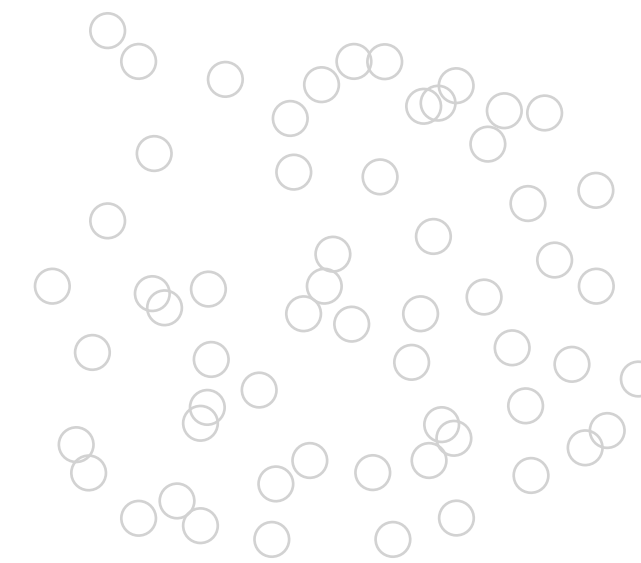
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Random (70/30)
(70/30 | k-fold)



Bands
(latitudinal, longitudinal)

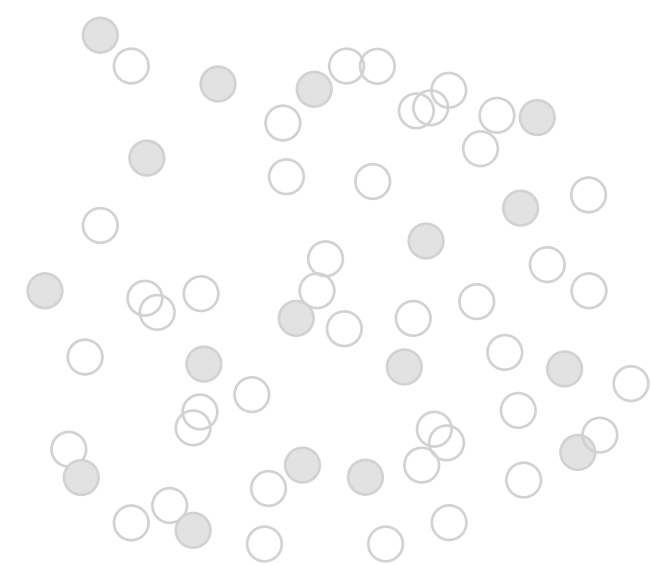


Blocks
(latitudinal, longitudinal)

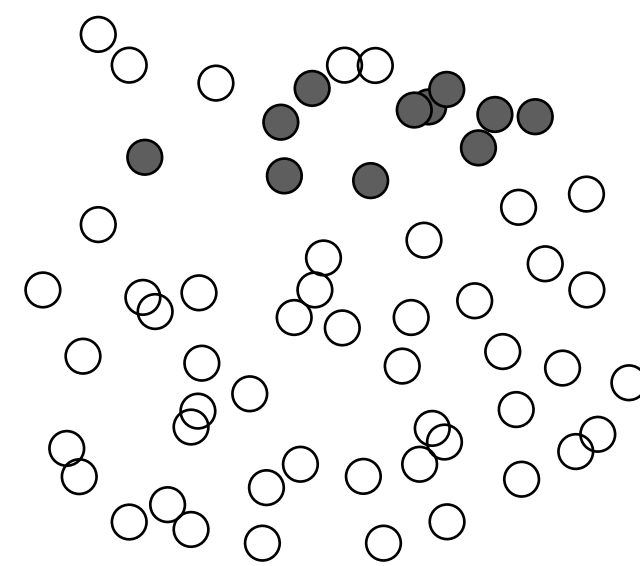
○ Training data ● Testing data

Missing independent data

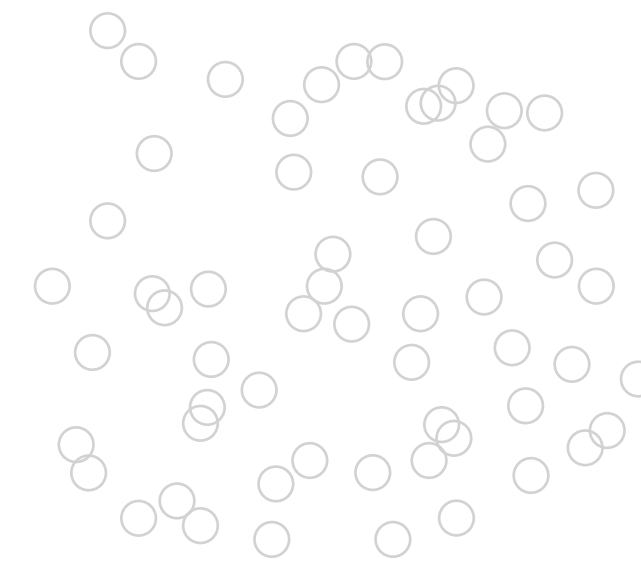
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Random (70/30)
(70/30 | k-fold)



Bands
(latitudinal, longitudinal)

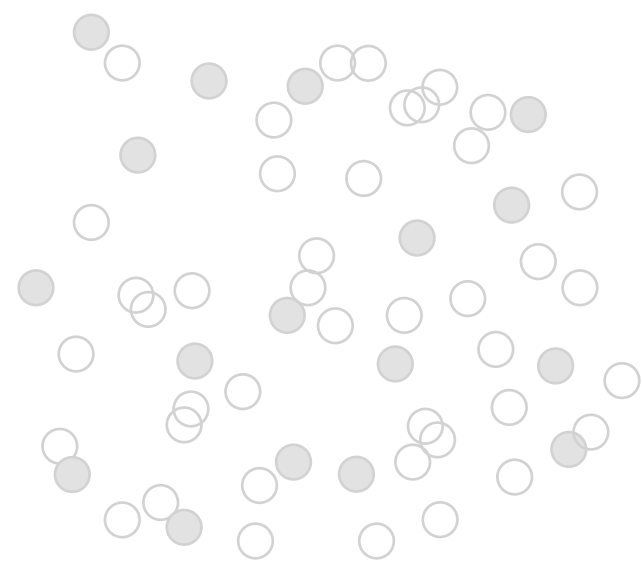


Blocks
(latitudinal, longitudinal)

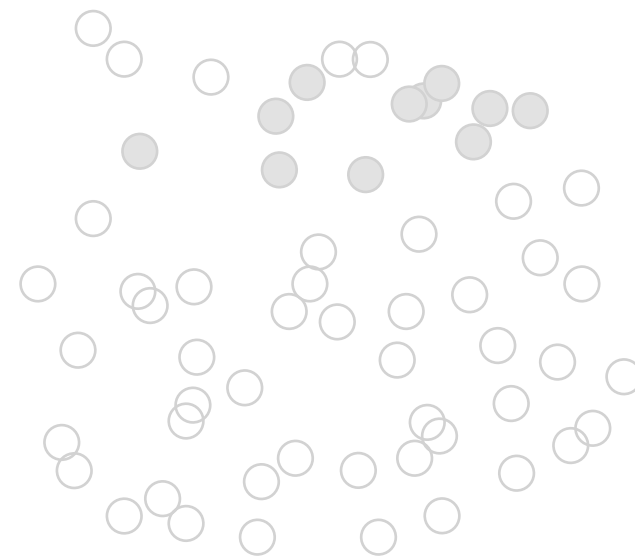
○ Training data ● Testing data

Missing independent data

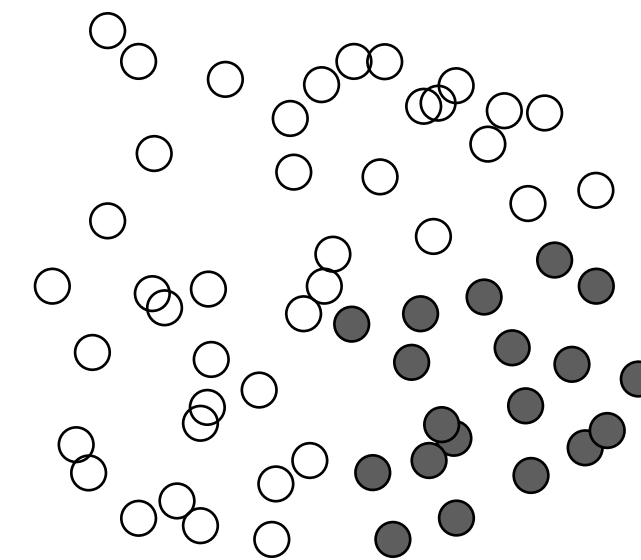
There are different methods to produce independent datasets. **Some approaches provide more independent datasets than others.**



Random (70/30)
(70/30 | k-fold)

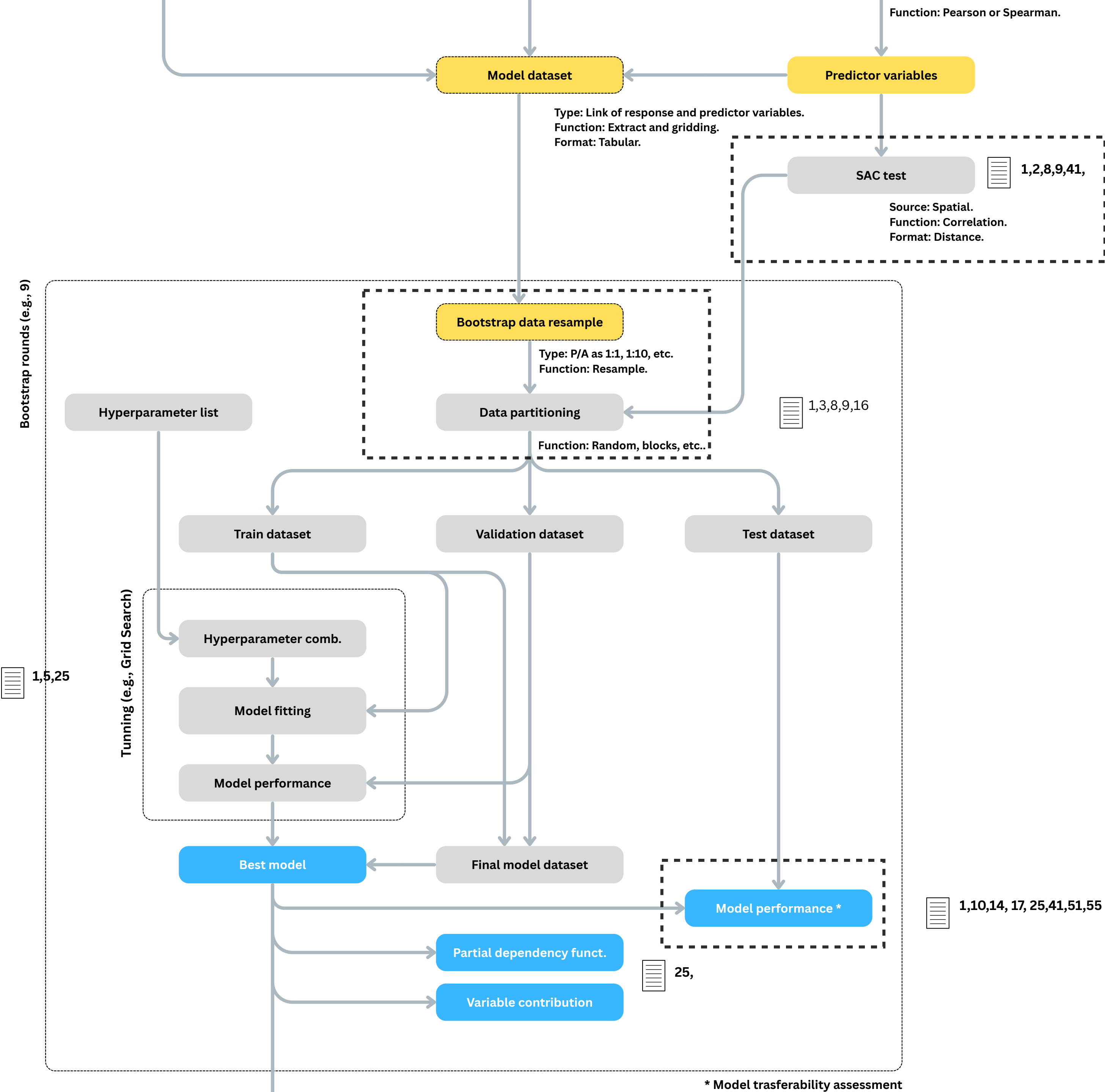


Bands
(latitudinal, longitudinal)



Blocks
(latitudinal, longitudinal)

○ Training data ● Testing data



Ensuring Ecological Plausibility

Beyond metrics: ecological realism

Does the model make biological sense?

Reporting metric scores (AUC-PR, TSS) is necessary but NOT sufficient!

Ecological plausibility check:

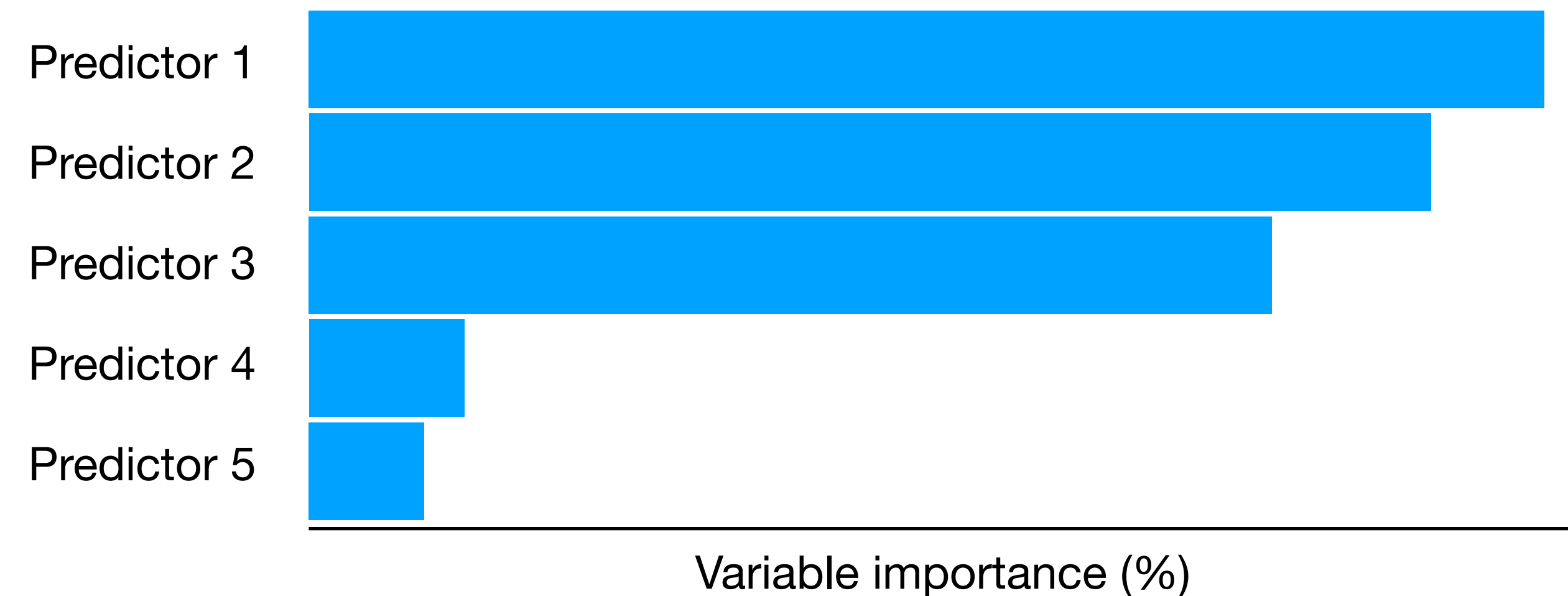
Do predictions align with known species biology, and habitat preferences?

Are the most important predictor variables ecologically relevant?

Relative variable importance

Assists in understanding the contribution of each variable to the model.

Approach: fit models with and without each variable, in order to determine the potential increase in performance. Without an important variable, a model should reduce its performance.

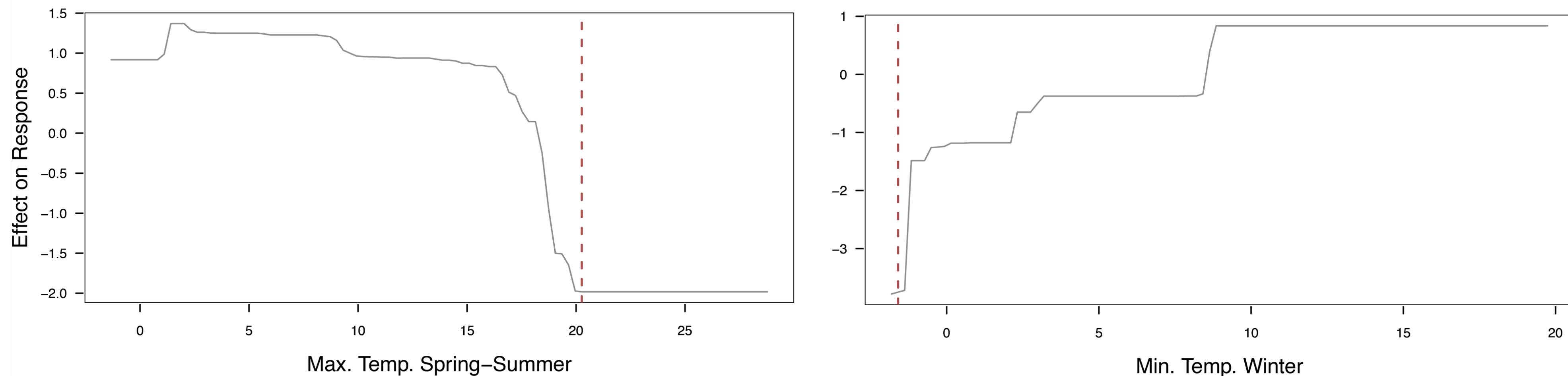


Does the model fit the expectations of ecological theory?

Partial dependency plots

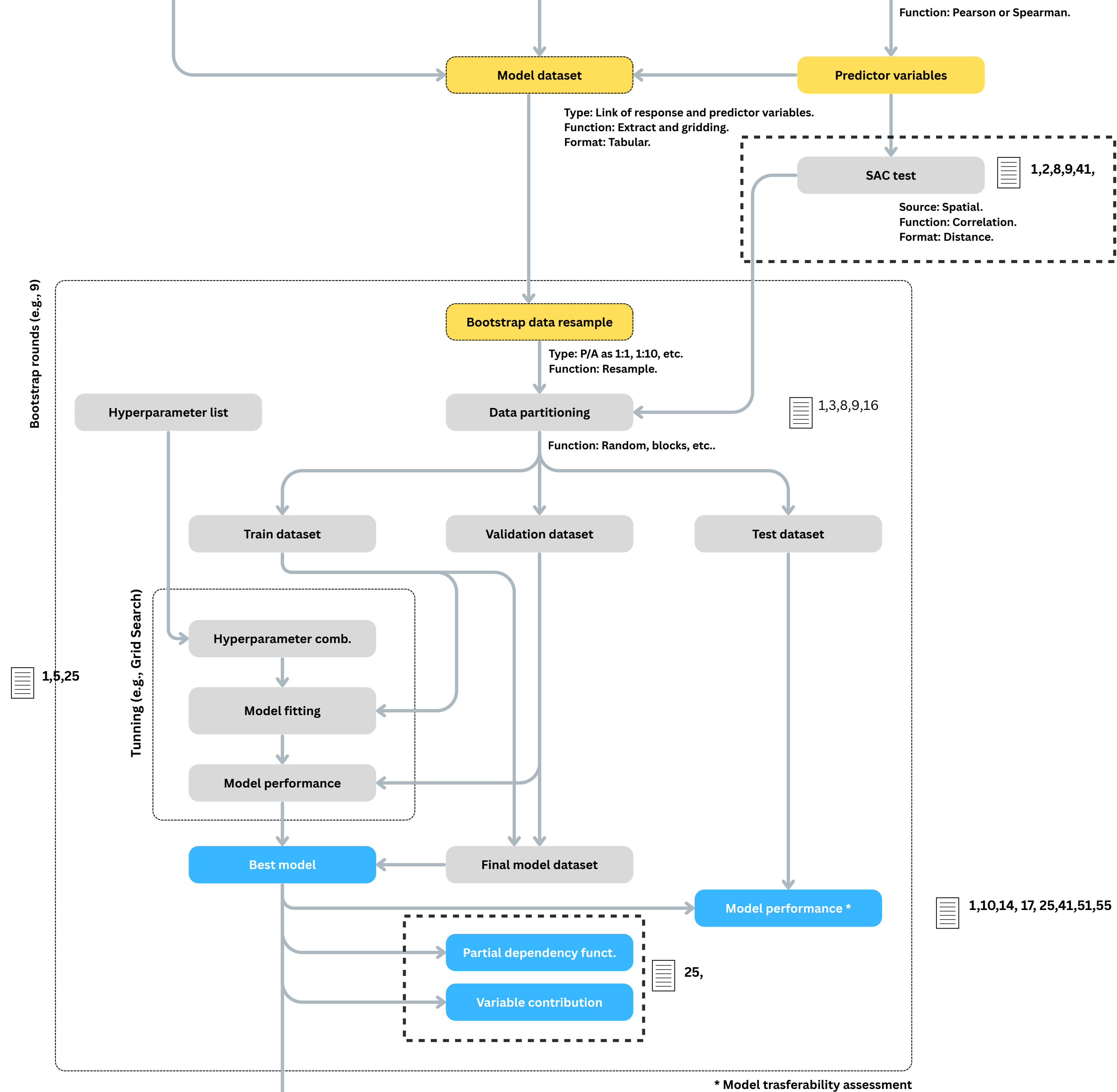
Show the relationship between the probability of occurrence and each variable.

Approach: the **response is predicted across the values of one environmental variable, while the other environmental variables are held constant at their mean.** The y-axis gives the impact on the response variable - the probability of occurrence.

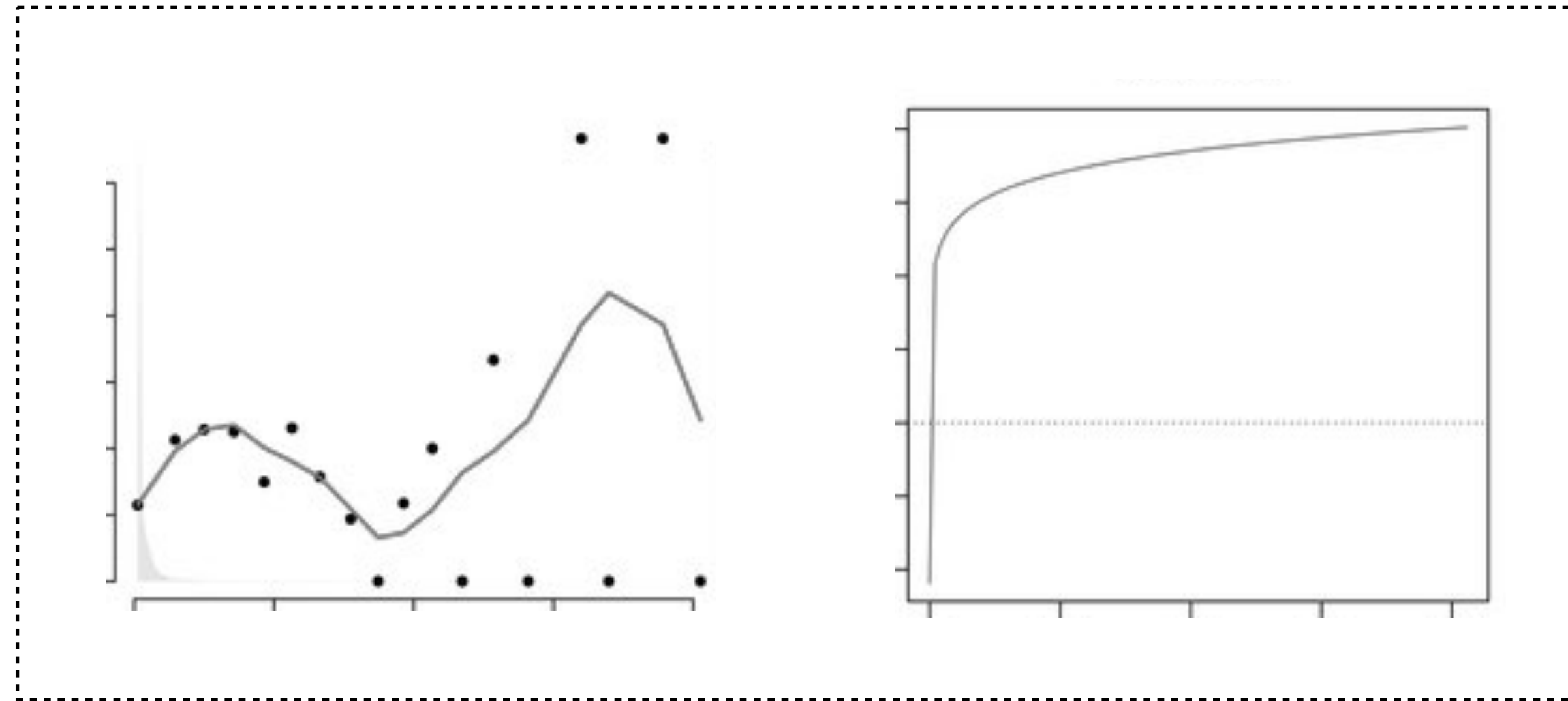


Does the model fit the expectations of ecological theory?

Curve and limiting points match physiological data (reliable model!)



Overfitted response vs. positive monotonicity



Forcing Ecological Rules (Monotonic Constraints)

Force the modeled relationship to follow a pre-defined, simple trend.

Prevents the model from fitting overly complex, potentially unrealistic shapes to the data for a particular variable.

The choice of monotonicity constraints is made *a priori* per variable **based on ecological theory** (no monotonicity, negative or positive monotonicity).

Only a few machine learning algorithms have such implementation (BRT; XGBoost).

Model Transferability and Environmental Novelty

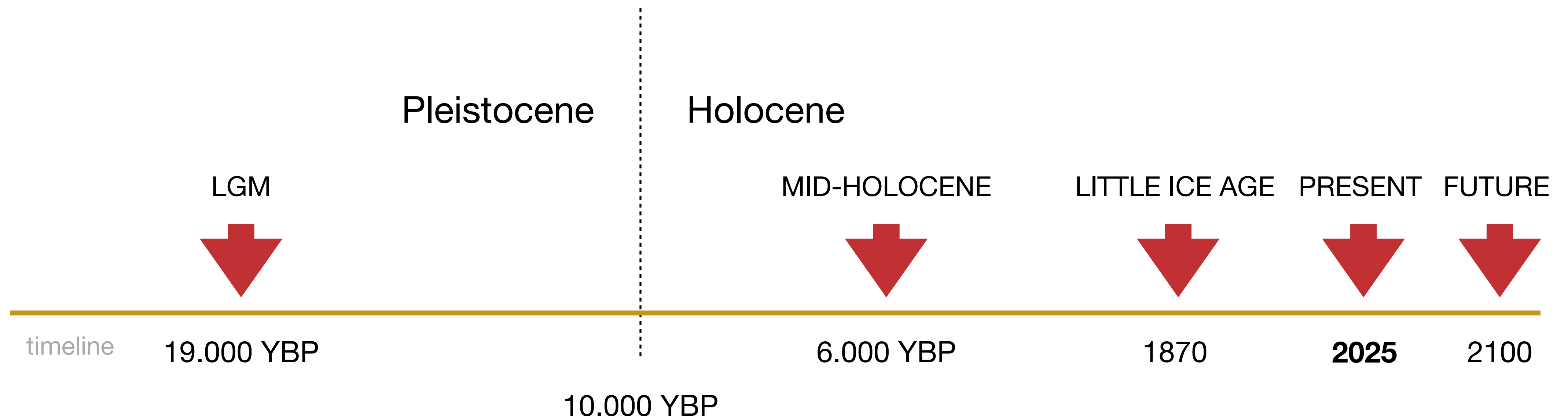
Model transferability

Ability of a model to predict beyond the data it was originally trained on.

Particularly **relevant when forecasting species distributions** under **climate change**.

Mainly dependent on:

1. Adequacy of environmental and species occurrence data to train the models;
2. Reduced complexity of the models (e.g., number of variables);
- 2. Proper choice / tuning of model parameterization.**

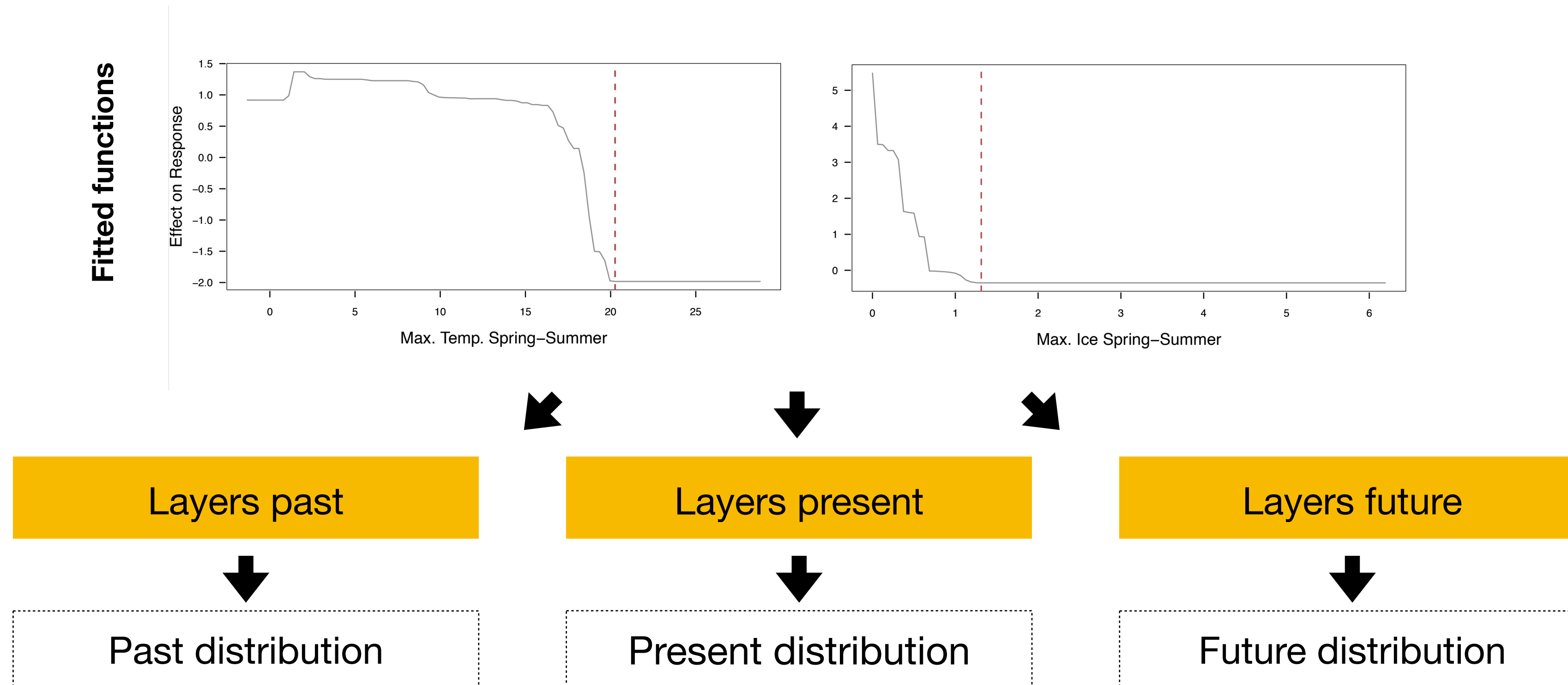


Predictive model-based transferability

Ability of a model to predict beyond the data it was originally trained on.

Process: a model fits **observed species occurrence data against environmental variables for the corresponding period**. The **same model is then applied directly to new climate data** representing new environmental conditions.

Predictive model-based transferability

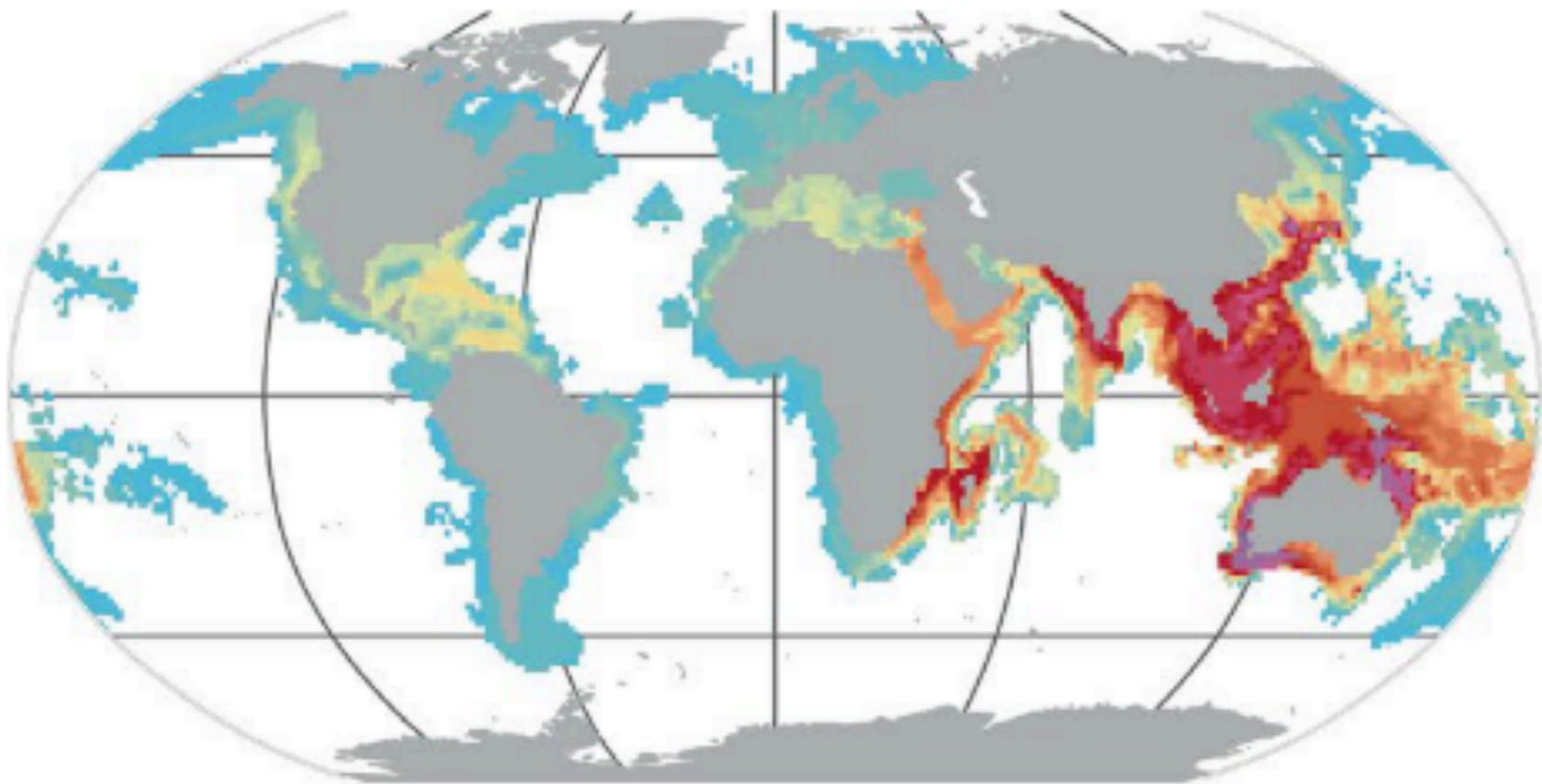


Regardless of the time period / region, **all environmental layers must be included in the transferability process.** So, the **availability of layers for climate scenarios also determines the choice of predictors for model fitting.**

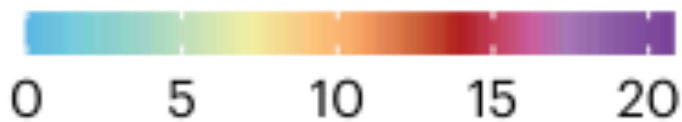
Incorrect SDM

Correct SDM

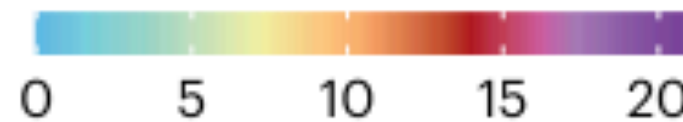
Present-day



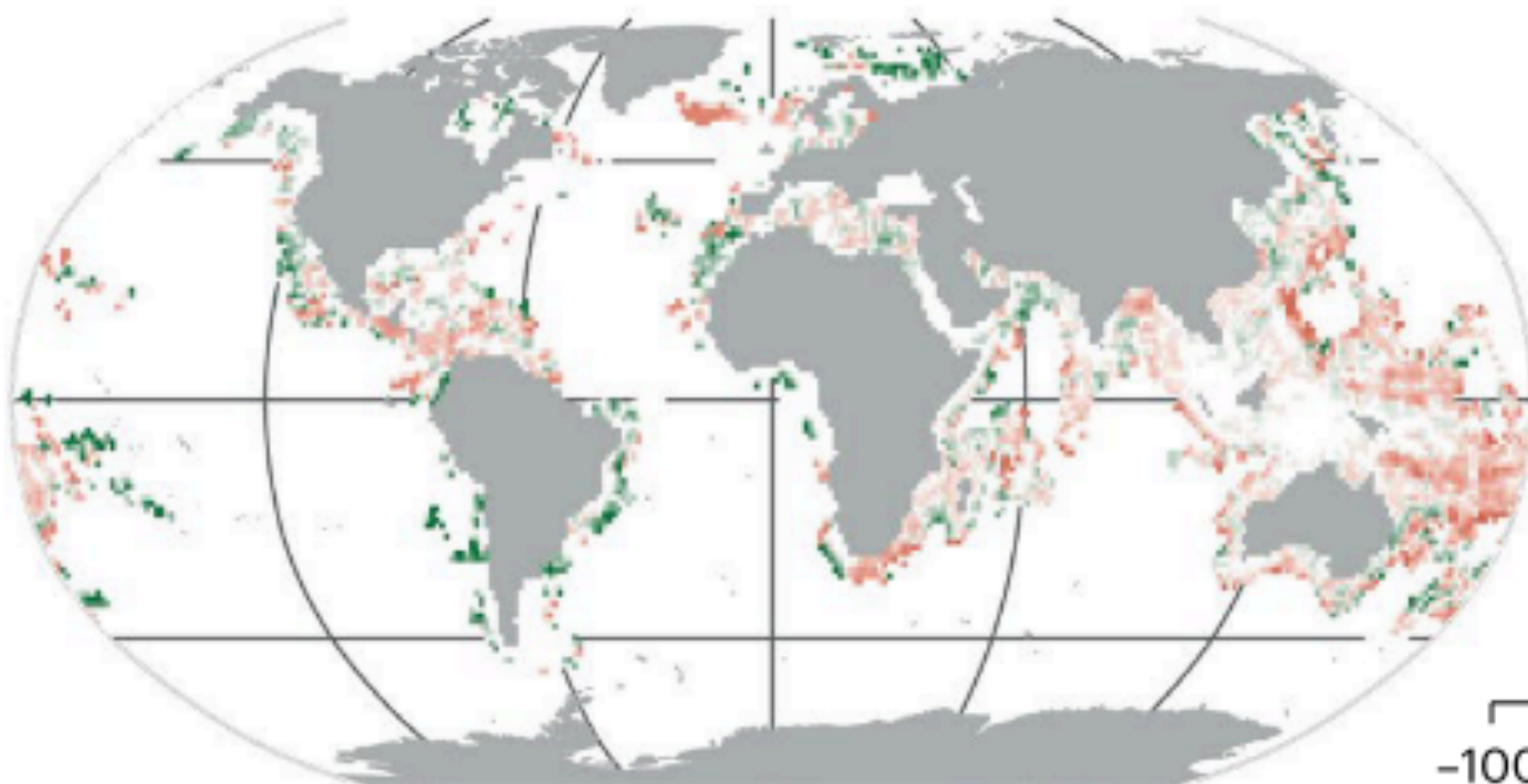
Species richness (no. of species)



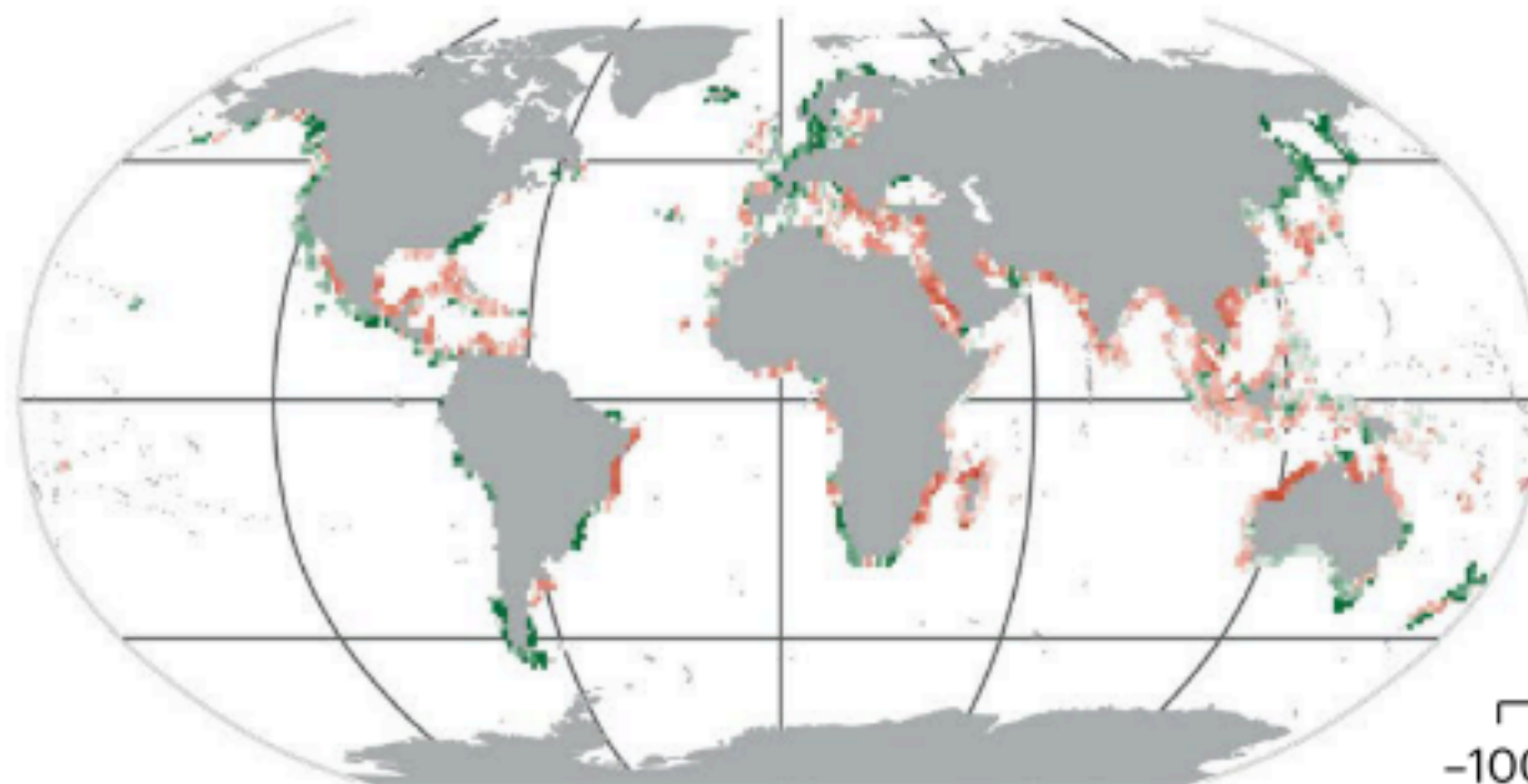
Species richness (no. of species)



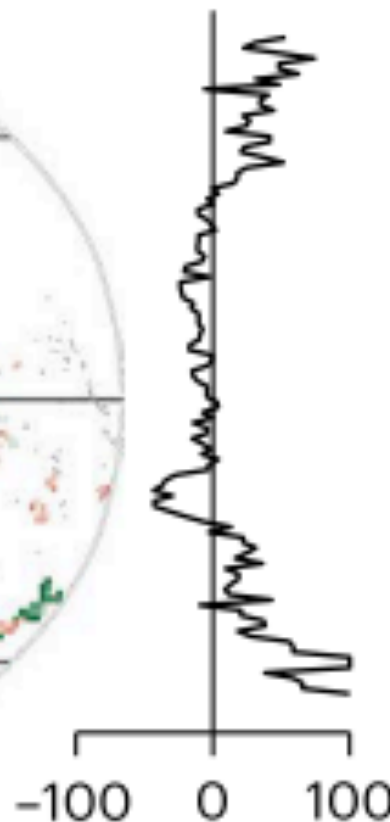
RCP 8.5 – present



Difference (%)



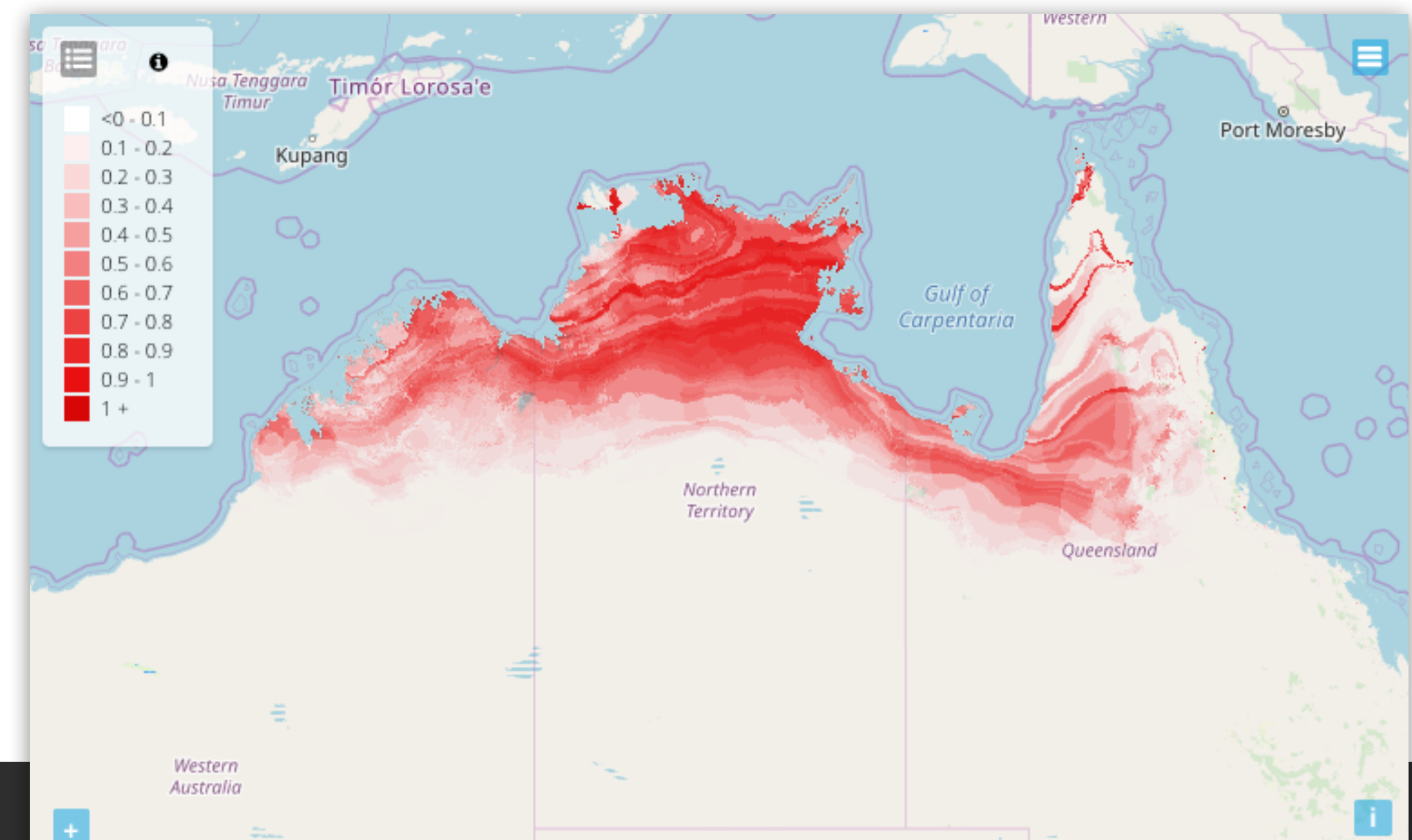
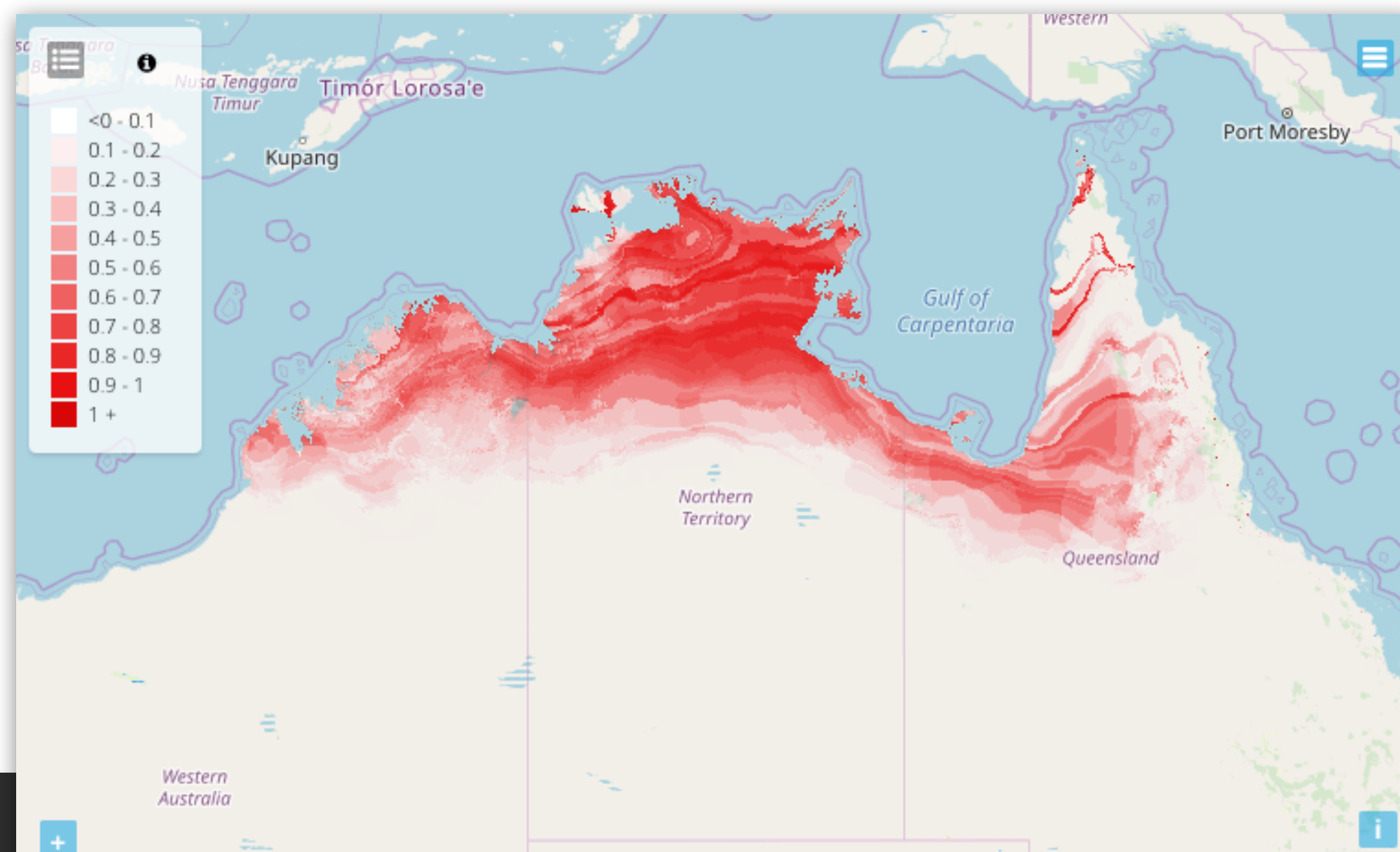
Difference (%)



Predictive model-based transferability

Model transferability can be analyzed (per cell) with different approaches:

(1) comparing **predictions** (e.g., probability) under different conditions.

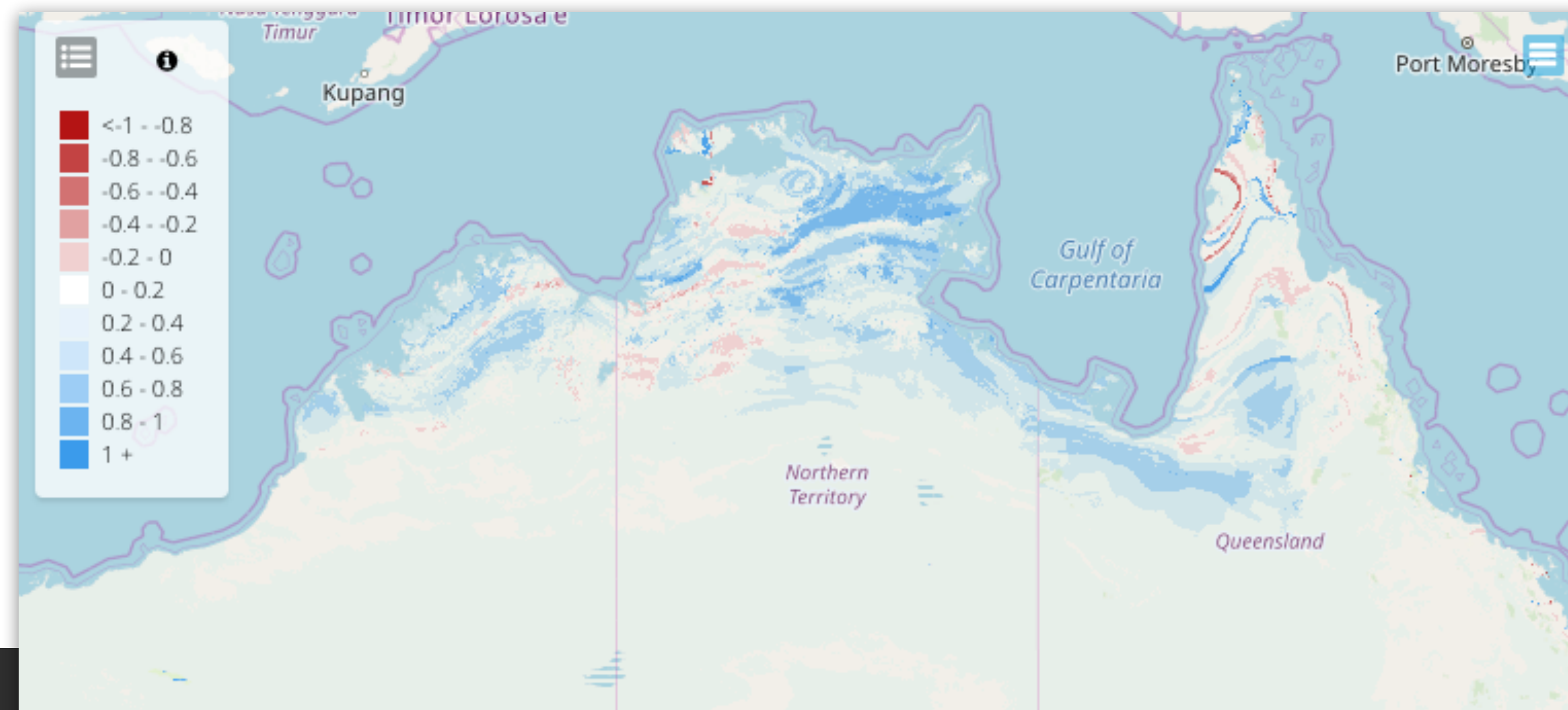


(2) **change in probability**, i.e., the difference in the predicted probabilities of different conditions;

e.g.,

[Layer 1 Future probability] - [Layer 2 present-day probability]

The map scales from -1 to 1, where negative values refer to lower suitable conditions and positive higher suitability.



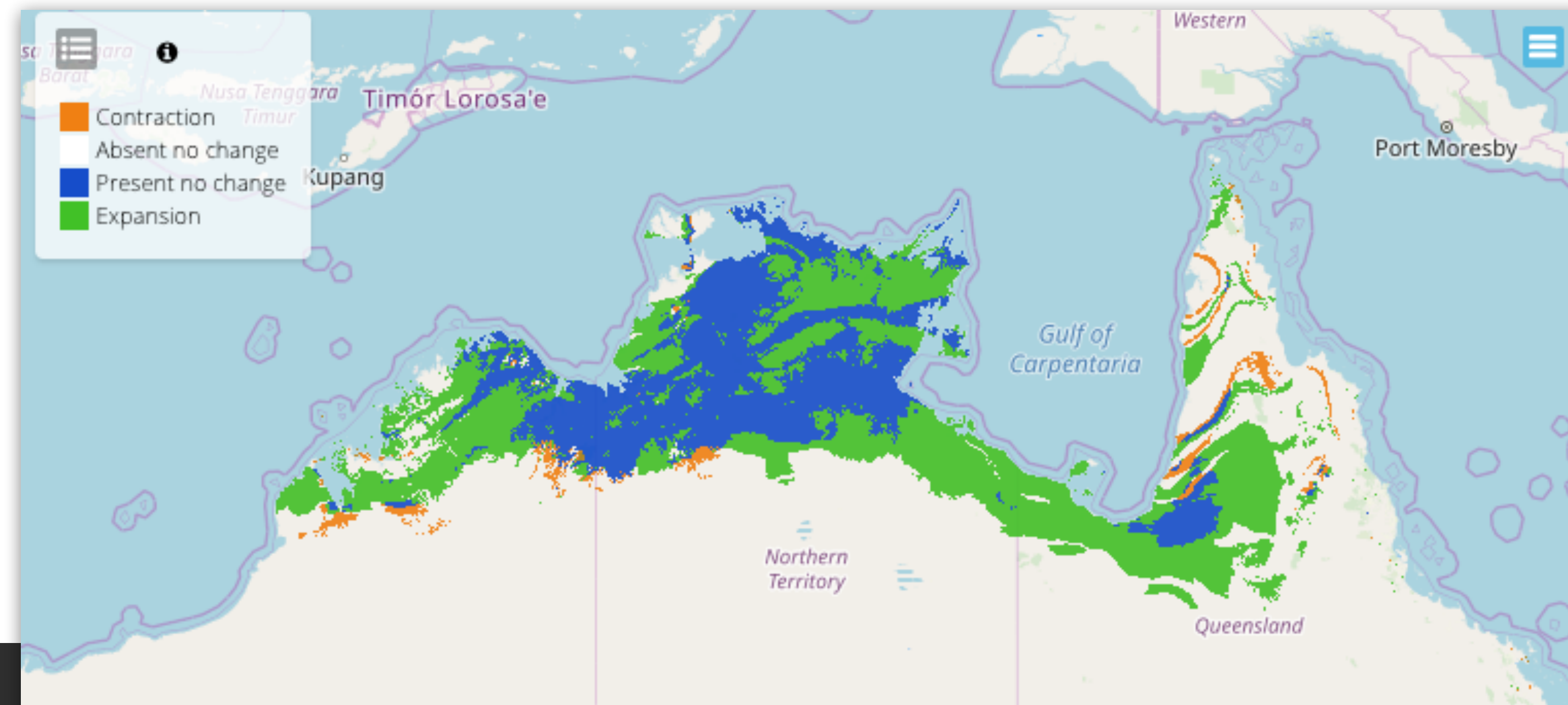
(3) **change in species distributional range**, generated with binary maps (i.e., 0 or 1).

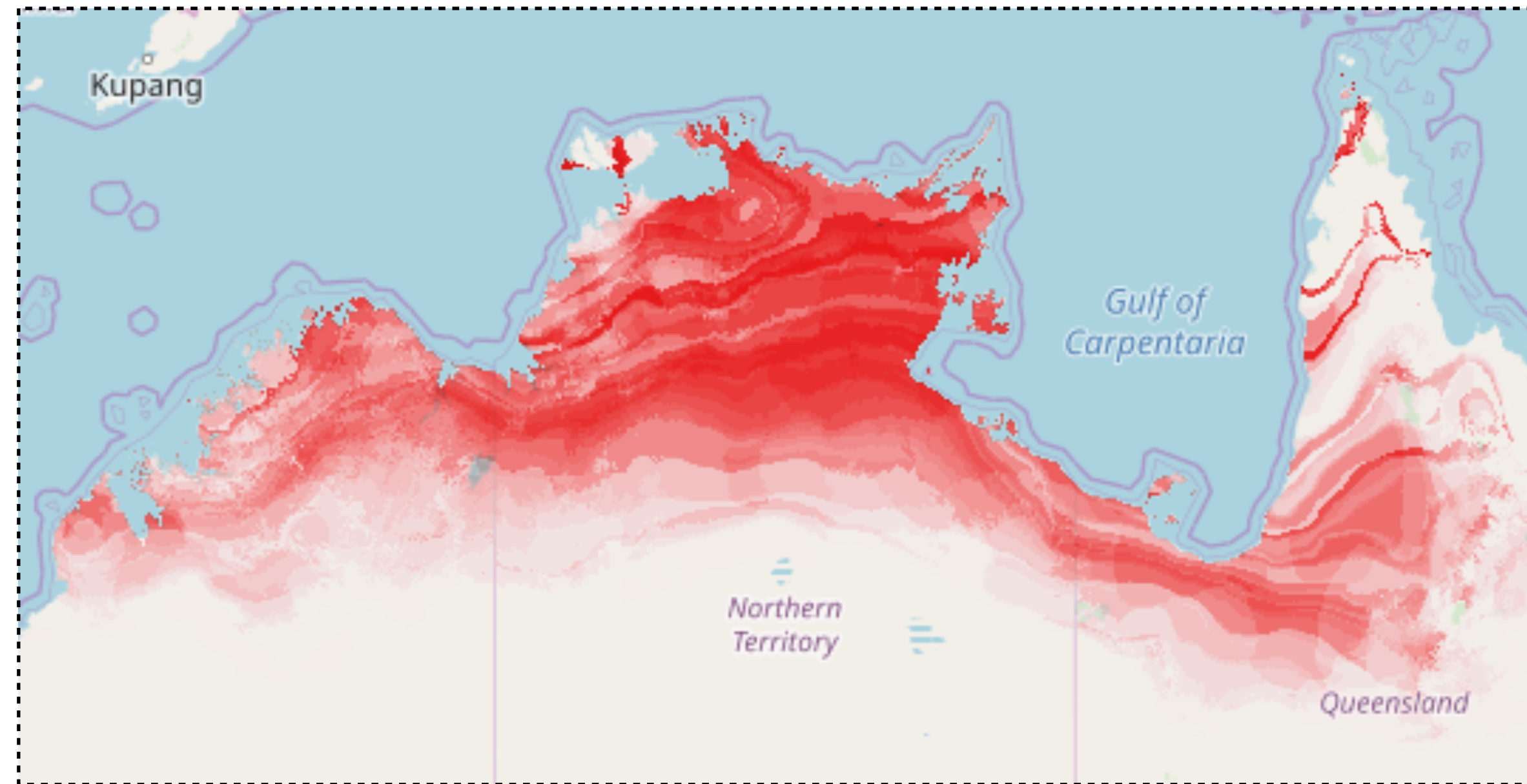
Comparing maps can indicate:

No change: present distribution equals future distribution;

Range loss or contraction: presence in the present and absence in future;

Range expansion: absence in the present and presence in the future.



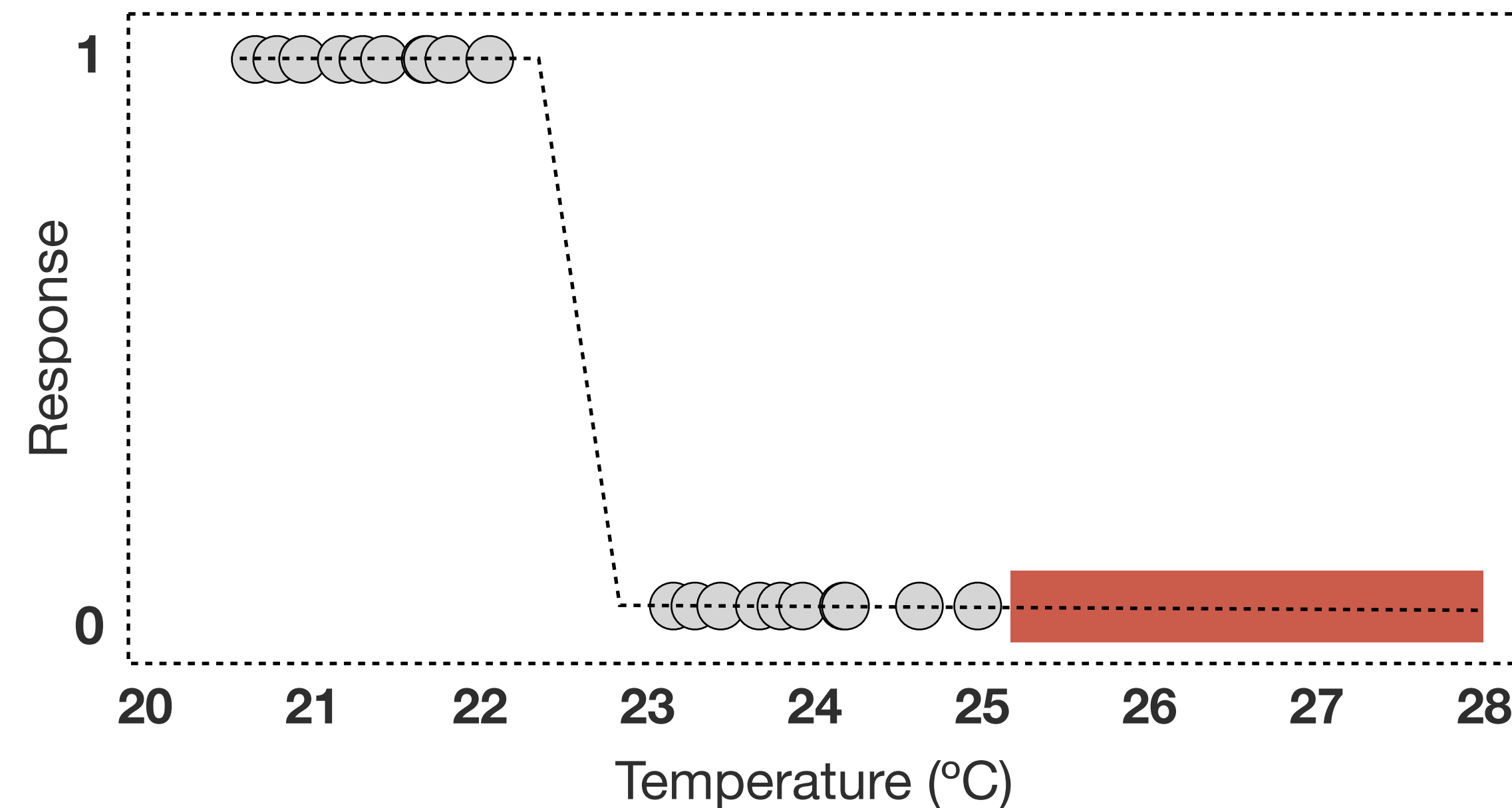


Predictive model-based interpolation

Predictions to **new sites within the range of values of environmental conditions sampled in the training data.**

From **scattered presence records to continuous distribution surfaces.**

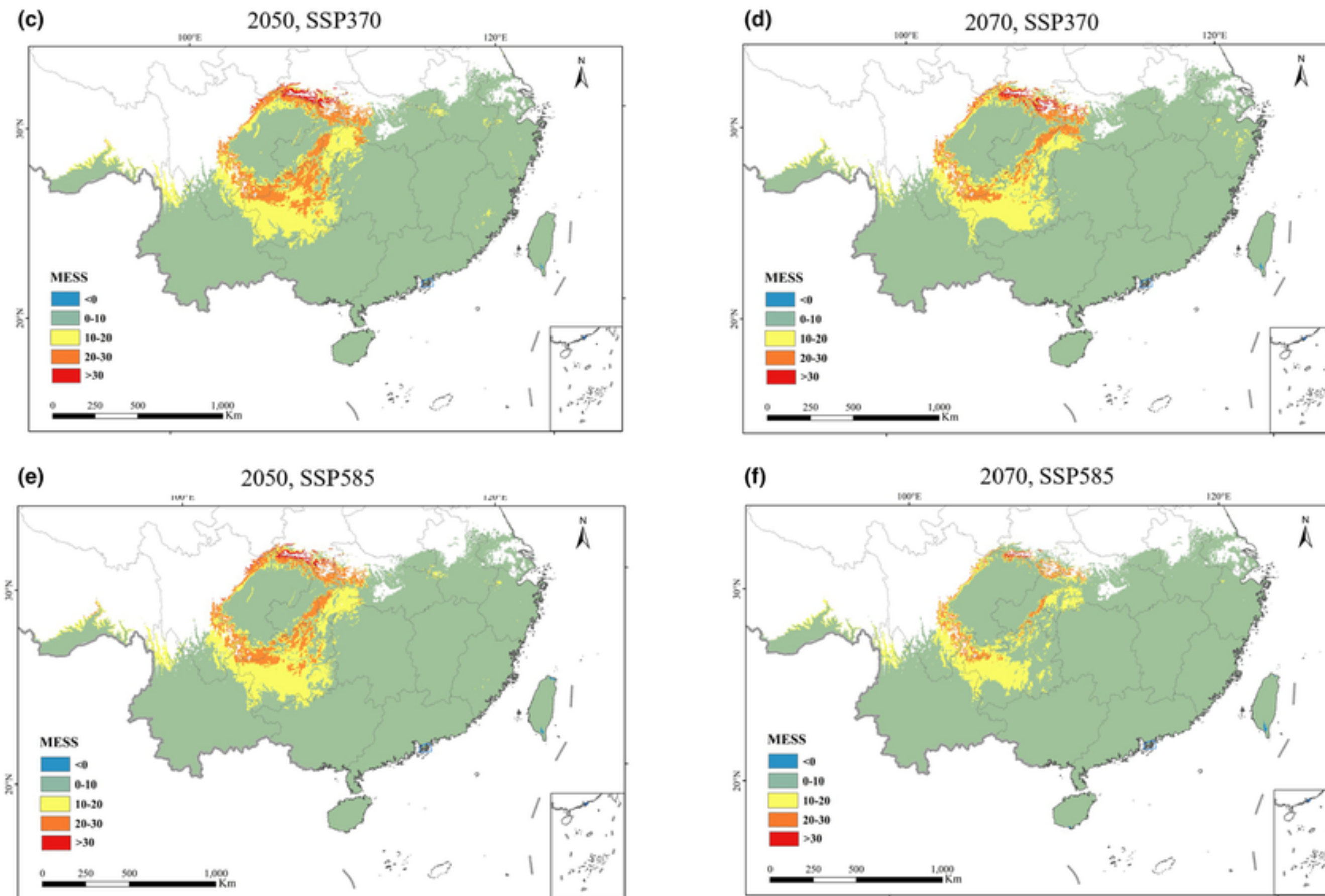
Maps do not show where species occur, but rather the distribution of regions with suitable environmental conditions.



Model transferability may lead to extrapolation, i.e., predictions to environmental values outside the range of environmental values used to fit the model.

e.g.,

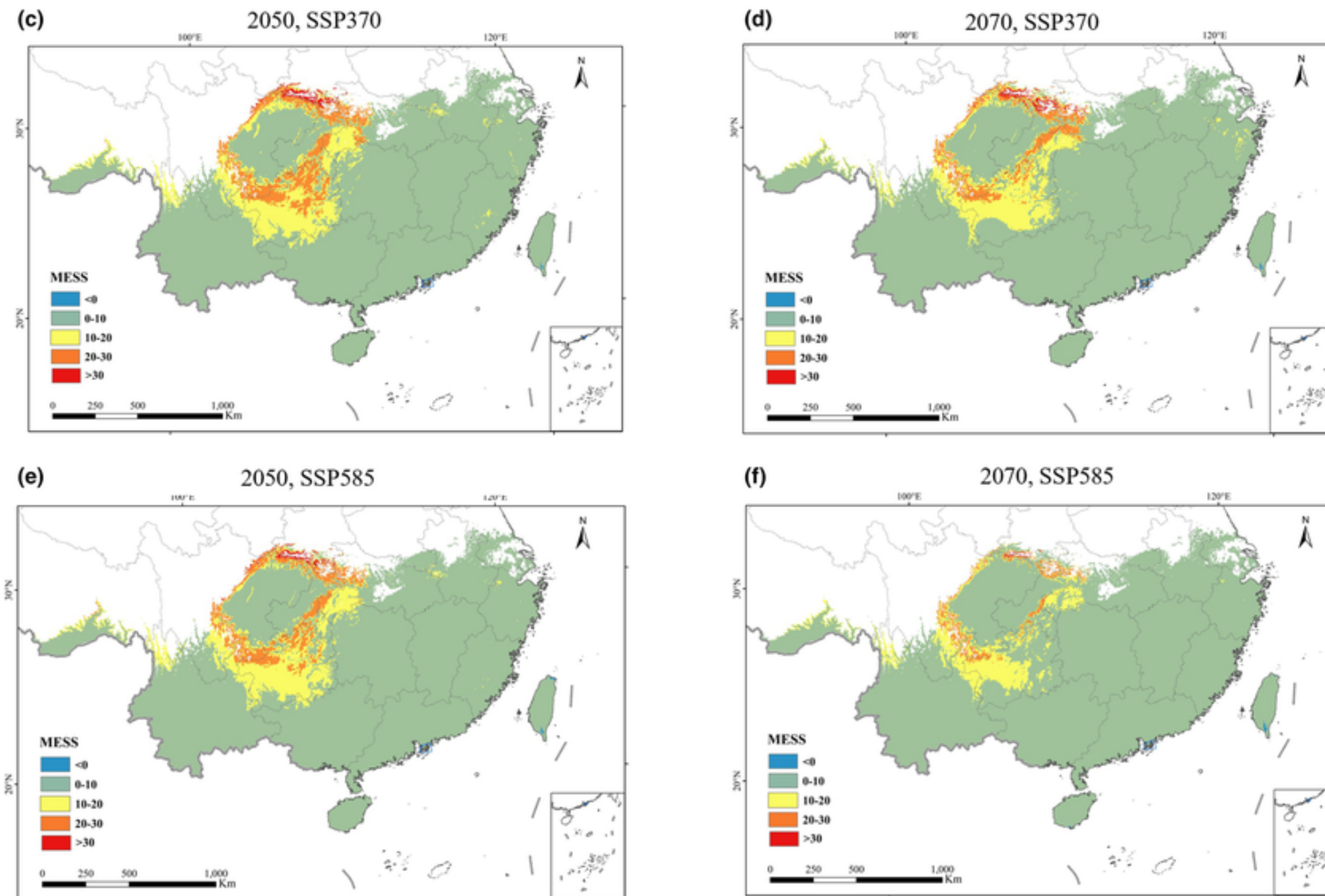
A model fitted records with temperatures of 20-25°C. If predictions are made for temperatures > 25°C, then the model will extrapolate. No information exists on the probability of occurrence at > 25°C, so the predictions will be uncertain.



Quantifying Environmental Novelty

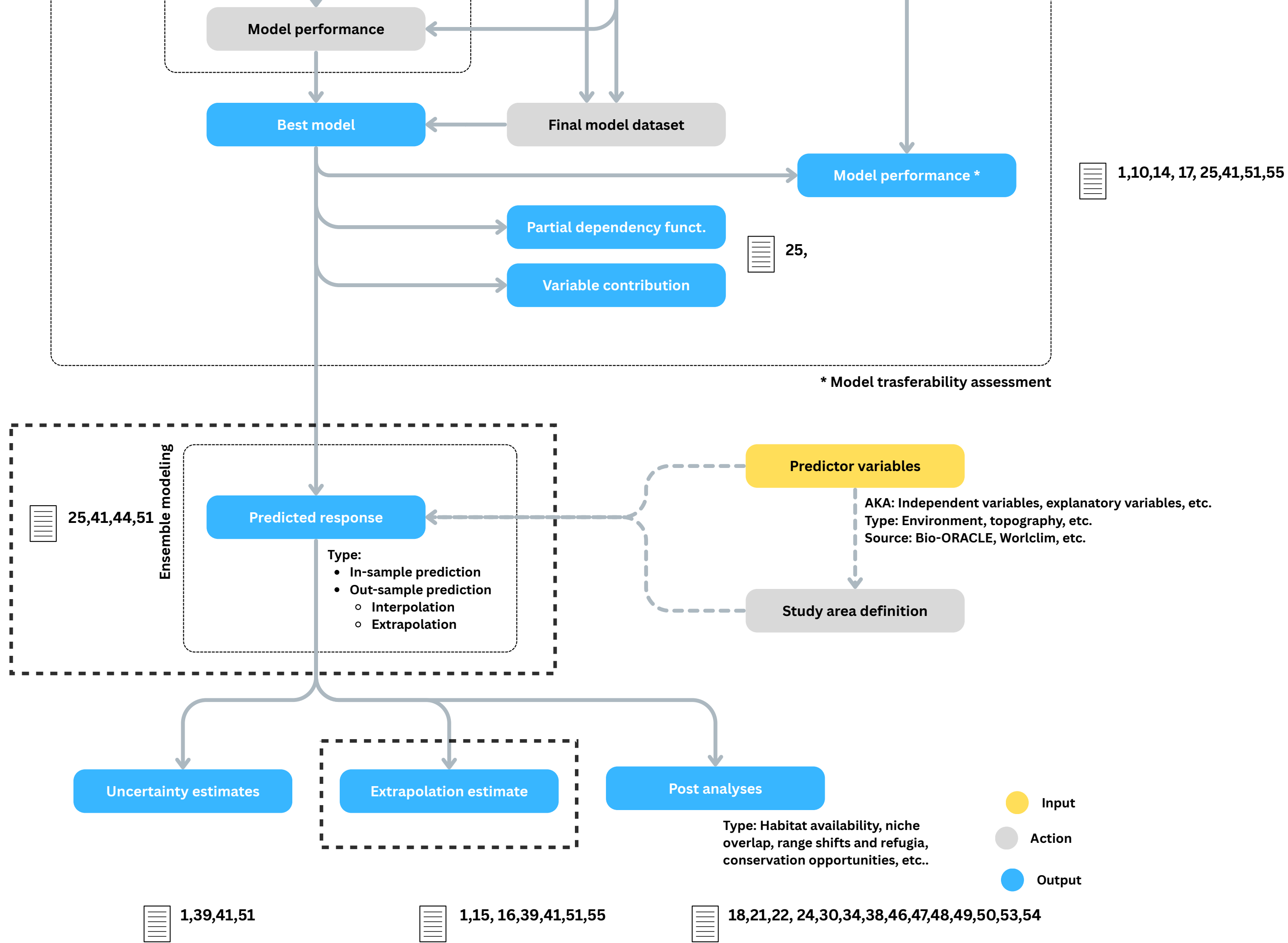
Future climates or new regions often contain conditions the model has never seen.

Multivariate Environmental Similarity Surfaces: A technique to map how environmentally similar (or dissimilar) the projection area is compared to the calibration area.



Positive Values (0 to 100): Similar Environments

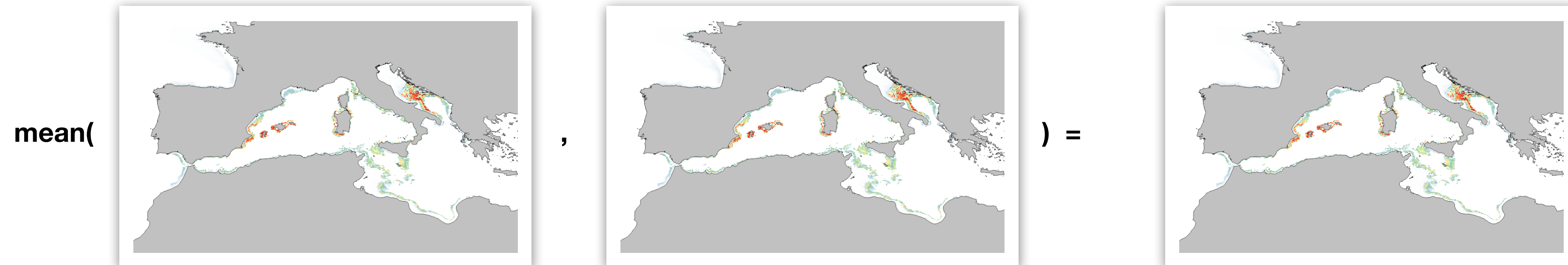
Negative Values (< 0): Novel Environments (Extrapolation)



Uncertainty estimates and Ensemble Modeling

Ensembling models

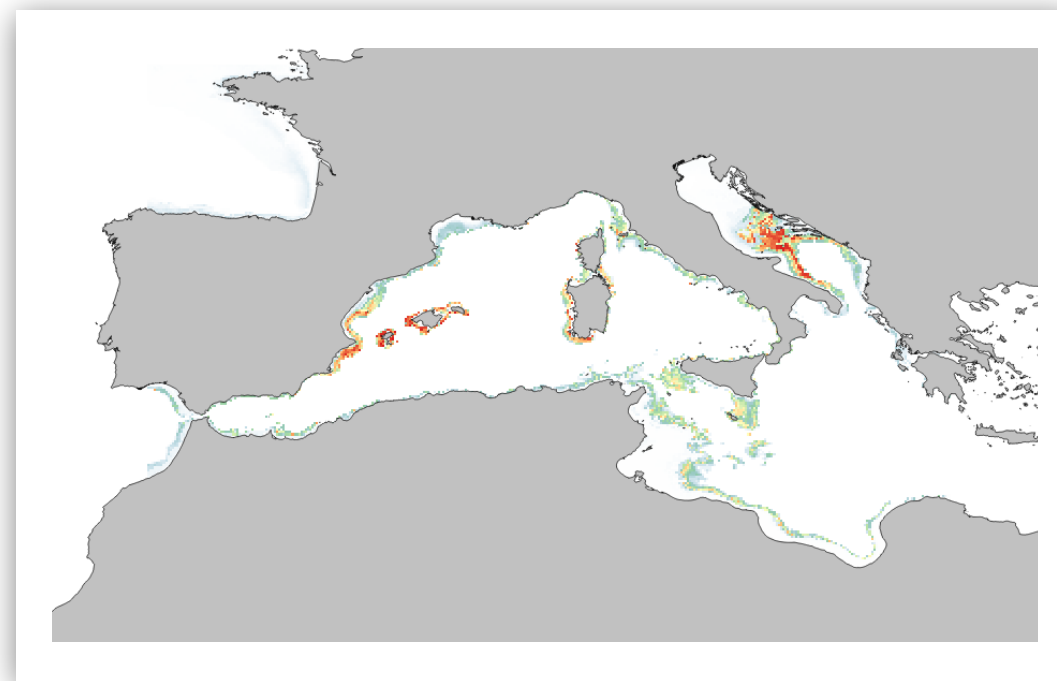
Instead of using a single algorithm to investigate species distributions, one can conduct an **ensemble experiment**. This **reduces the uncertainty of algorithms by combining their outputs with a general statistics** (e.g., mean).



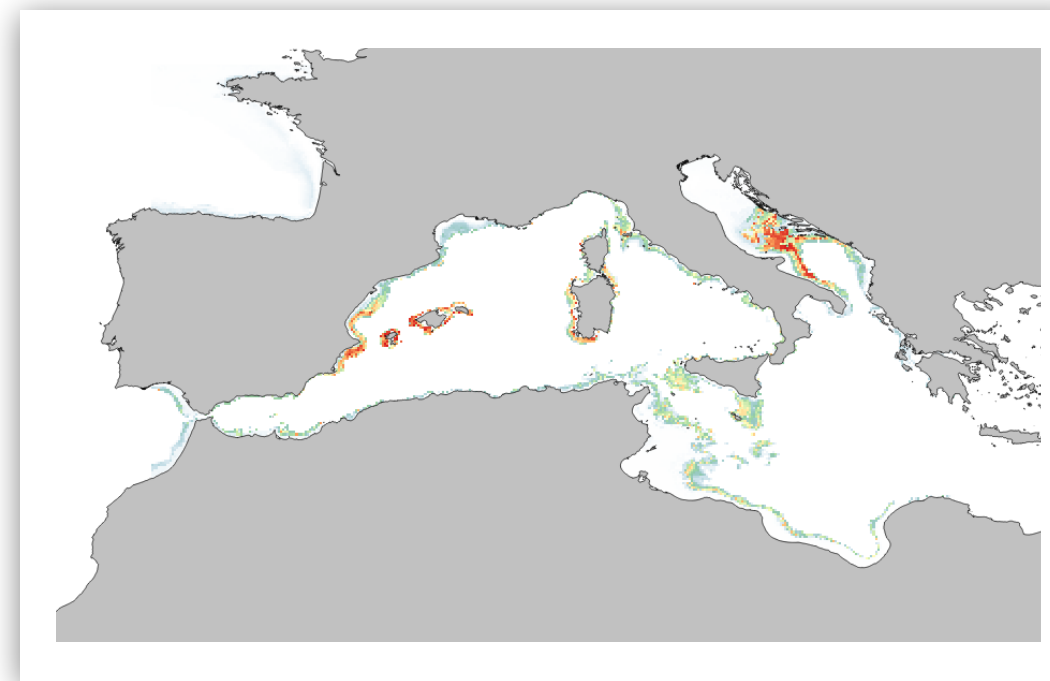
Mapping uncertainty between models

Ensembles allow us to map where different algorithms agree and where they heavily disagree. By mapping the standard deviation between the models, we can visually identify high-confidence habitats versus highly uncertain regions

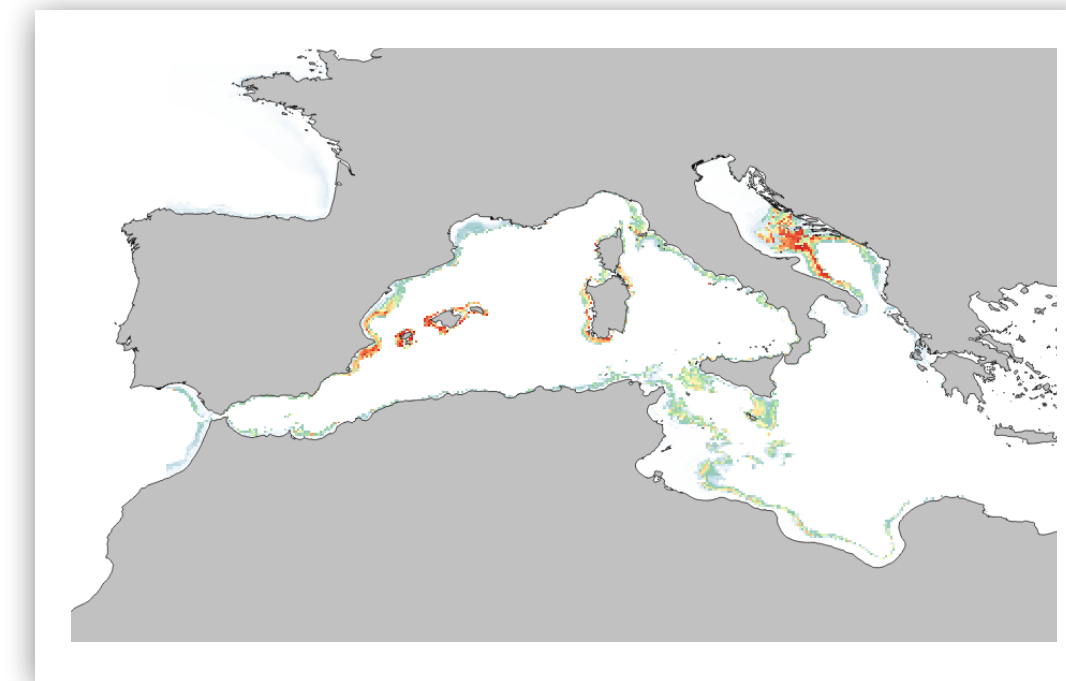
SD(



,



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References

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Zurell et al., 2020: <https://doi.org/10.1111/ecog.04960>

Barbet-Massin et al., 2012: <https://doi.org/10.1111/j.2041-210X.2011.00172.x>

Fei & Yu 2015 : <http://dx.doi.org/10.1007/s10980-015-0272-7>

Elith et al. 2009: <https://doi.org/10.1146/annurev.ecolsys.110308.120159>

Assis et al., 2024: <https://doi.org/10.1038/s41477-024-01735-7>



Performance of algorithms in SDM

Most algorithms have been tested and compared.

Differences in performance among different algorithm types tend to be smaller than differences among species.

Machine learning algorithms, like MaxEnt, and especially those that incorporate model averaging (e.g., BRT and XGBoost) tend to have **better performance than more simple statistical methods** such as GLMs.